

AGEL J020613–011417 — Drafting Fact Sheet for the Methods & Results Sections

Facts, decisions, and provenance keyed to the paper outline — last revised 2026-06-12
(lens-model summary from Ferrami draft; post-M12)

Compiled for R. Córdova Rosado from project notes, memory, METHODS_AND_SYSTEMATICS.md, and TESTS_AND_DIAGNOSTICS.md

2026-06-08

How to use this document

This is a drafting reference, not prose for the paper. Every bullet is a fact or decision we have already made, with its source file in [brackets] so you can verify it. Where a number is superseded, the current value is given and the stale one flagged. The `<!-- PV:auto:... -->` blocks (the headline numbers below + the §3.1 budget table) are generated from `results/PAPER_VALUES.json` — do not hand-edit inside the markers; refresh with `python scripts/paper_values.py --render DRAFTING_FACTS_paper_2026-05-29.md`. Three gaps are flagged up front — they are not in this repo and you must source them elsewhere before drafting those paragraphs:

- G1 — Lens modeling (Ferrami et al.) — RESOLVED 2026-06-12 (provisional). Summarised into §2.3 (method) and §3.3 (results) from the Ferrami et al. June-12 draft (`7Documents/AGEL/AGEL_0206_ApJL_Figures/SMBH_in_Einstein_Spiral_DESJ0206_0114_June12/main.md`): pronto code, sEPL/cEPL/bEPL halos, point-mass vs SPS/NFW perturber test, 15-model evidence table, best-model parameters, $M\text{--}\sigma$ placement. All numbers PROVISIONAL (draft stamps them PLACEHOLDER UNTIL FINAL RUNS) and on Planck-2015 cosmology ($\neq$ our $H_0=70$) — see §2.3 flags. Lens-modeling repos remain `7Documents/AGEL/202509_DESJ0206_modeling/`, `20251112_DESJ0206_Pyauto_PRONTO/` for the underlying runs.
- G2 — X-ray / “truly quiescent” statement — PARTLY RESOLVED 2026-06-12. The Ferrami draft §Data supplies the quiescence argument (Chandra DR2 X-ray flux limit $1.8 \times 10^{-15} \text{ erg s}^{-1} \text{ cm}^{-2} \rightarrow L_X < 10^{43} \approx 10^{-4} L_{\text{Edd}}$; LOFAR DR3 radio-quiet; no AGN lines in KCWI) — captured in §3.3.6. Still verify the Chandra/LOFAR source values directly before asserting them in *our* paper.
- G3 — “We do not account for peculiar velocities.” This sentence is not currently written in any methods doc. It is a true statement about our analysis (we use the line-fit systemic z directly) — just add it; nothing to reconcile.

Headline numbers — *generated from results/PAPER_VALUES.json by scripts/paper_values.py --render; do not hand-edit inside the markers.*

- $\sigma_e(<R_e) = 267.31 \pm 11.77 \text{ km/s}$ (asym $-11.98 / +11.58$); often rounded $267 \pm 12 \text{ km/s}$. Measured at the best-mask $R_e=2.097''$ aperture; the R_e -source systematic (± 5.13 , best-mask CoG family) is folded into the budget (§3.1).
- $\log(M_\star/M_\odot) = 11.46 + 0.07 / -0.06$ (stat, 10%) — aperture-corrected total at the matched $2 R_e = 4.19''$ elliptical aperture (empirical aperture + single-Sérsic model wings; GAMA/Taylor+2011 fluxscale). Five estimators: raw 11.27 (empirical) · raw+apcorr 11.33 · filled 11.42 · total 11.46 (headline) · Sérsic-total 11.49. Sérsic-only systematic ± 0.19 dex; masking-approach ± 0.09 dex. See §2.1.1b / §3.2.
- $R_e = 2.097'' = 15.26 \text{ kpc}$ (best-mask F140W+F200LP CoG; method systematic $\pm 0.10''$). Supersedes 2.305'' (expert mask).
- $z_{\text{deflector}} = 0.67564$; $z_{\text{source}} = 1.303$.

Cosmology (Planck 2015 — adopted 2026-06-12 to match the companion lens-modeling paper, Ferrami et al.): FlatLambdaCDM, $H_0 = 67.7$ km/s/Mpc, $\Omega_m = 0.302$; $1'' \approx 7.2764$ kpc, $D_A = 1500.9$ Mpc, $D_L = 4214.1$ Mpc at $z = 0.67564$. (*Was $H_0=70$, $\Omega_m=0.3 \rightarrow 7.04$ kpc / $D_A=1452$ Mpc before this date.*) [reference_hst_jwst_data_properties.md]

Cosmology switch — impact (2026-06-12). Matching Planck 2015 changes only the distance-scaled quantities, and only mildly (D_L and D_A both $\times 1.0331$ at our fixed $z_l = 0.67564$): - R_e : $14.76 \rightarrow 15.26$ kpc (+3.3%). R_e in arcsec ($2.097''$) is unchanged. - $\log M_\star$: $11.47 \rightarrow 11.50$ (+0.0282 dex). This is an exact D_L^2 rescale of the flux-calibrated SED mass (the M/L is color-driven, hence cosmology-independent), so it is a uniform additive shift of *every* estimator — no Bagpipes re-fit needed. Well inside the ± 0.09 (stat) / ± 0.19 (Sérsic sys) budget $\rightarrow$ not a dominant systematic. - $\sigma_e = 267.31 \pm 11.77$ km/s and every angular quantity (θ_E , R_e , masks, apertures) are IDENTICAL — velocities and angles do not depend on cosmology. - Implementation: the PV:auto headline + §3.1 budget are re-rendered under Planck 2015 (scripts/paper_values.py: KPC=7.2764; DLOGM_COSMO=+0.0282 applied to all $M_\star$ estimators). TODO: drop DLOGM_COSMO once results/aperture_matched_photometry.npz is regenerated under Planck 2015 upstream, else it double-counts. - Redshift reconciliation — TODO (sub-dominant). Ferrami quotes $z_l = 0.675$ / $z_s = 1.303$ but takes the redshifts FROM us; our line-fit $z_l = 0.67564$, $z_s = 1.30263$ (notebook 04) are canonical and should propagate into their final runs. The $\sim 6 \times 10^{-4}$ / 1×10^{-3} differences shift D_A by $< 0.1\%$ — not a dominant systematic, left as a to-do.

Staleness warning: older docs (METHODS_AND_SYSTEMATICS.md 2026-05-18 body; pre-2026-06-11 handoffs) print superseded σ_e numbers (254.85 / 268.98 / 269.62, totals $\pm 18/\pm 13.27$). The current headline is 267.31 ± 11.77 (post “best mask throughout”, 2026-06-11, $R_e=2.097''$). Always quote the current value (canonical source: results/PAPER_VALUES.json).

§1 Data and observations

All values below are read directly from the FITS headers of the analysis files (verified 2026-06-08), not from memory. Target: AGEL J020613–011417 = DES J0206–0114, ICRS RA = 31.55611° , Dec = -1.23817° (HST RA_TARG/DEC_TARG); deflector $z = 0.67564$, source $z = 1.302$. The field is the massive cluster ACT-CL J0206.2–0114 (ACT DR5, Hilton et al. 2021) — the JWST pointing target.

Facility	Inst. / mode	Band(s)	Program (PI)	UT date	Exp.	Pixel scale
Keck II	KCWI — KCRM red (RL) + KCB blue (BL), Medium slicer, 2×2	RL 5625–8941 Å (σ_e); BL λc 4500 Å	U204 + K409 + U002 (3-night combine)	2024-12 → 2025-11	≈ 180 min RED + ≈ 198 min BLUE (36 × 300 s RED)	0.300"/spaxel
HST	WFC3/UVIS	F200LP	16773 (Glazebrook)	2022-07-14	600.0 s (NDRIZIM 4)	0.050"/pix
HST	WFC3/IR	F140W	16773 (Glazebrook)	2022-07-14	597.7 s (NDRIZIM 3)	0.080"/pix
JWST	NIRCam SW, module B	F150W2 (CLEAR)	05594 (Mahler)	2024-08-27	1836.0 s	0.03075"/pix
JWST	NIRCam LW, module B	F322W2 (CLEAR)	05594 (Mahler)	2024-08-27	1836.0 s	0.06301"/pix

§1.1 Keck/KCWI IFU spectroscopy (the σ_e data)

- Keck II / KCWI, KCRM red arm, RL (Red-Low) grating (RGRATNAM), Medium slicer (IFUNAM), 2×2 binning (BINNING/CCDSUM); nod-and-shuffle off (RNASNAM= 'Open ').

- The headline cube is a 3-NIGHT, 3-PROGRAM combine (see §2.2.1), NOT a single dataset — the cube header (DATE-OBS 2025-11-17, PROGPI Jones, XPOSURE 300 s) describes only the first input frame, so cite the combine, not the header:

Night (UT)	Program (PI)	RED (RL)	BLUE (BL)
2024 Dec 29	U204	4 × 300 s	1 × 1320 s (<i>excluded by reducer</i>)
2025 Aug 30	K409 (PI TBD)	12 × 300 s	4 × 990 s
2025 Nov 17	U002 (Jones)	20 × 300 s	5 × 1320 s

→ 36 × 300 s ≈ 180 min RED + ≈ 198 min BLUE on-source. The NEW `_mtwdo_` reduction (Master-Twilight + Dome hybrid flats, K. R. Gupta), Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits; dithered (PA 0°/45°/90°); CR rejection via astroscrappy (fsmode=median, psffwhm=2.50). OBSERVER = Glazebrook, Gupta, Jones, Kacprzak, Alcorn, Rhoades, Barone, Tran, Chen, Vasa. (*K409 PI + Aug-30 DIMM seeing still TBD for acknowledgements.*)

- Both KCWI arms recorded simultaneously. The red arm (KCRM, RL grating, central λ 7150 Å RCWAVE — the cube below) is the σ_e dataset. The blue arm (KCWI/KCB, BL (Blue-Low) grating BGRATNAM, central λ 4500 Å BCWAVE, KBlue filter BFILTNAME, N&S off) is the ≈198 min BLUE above; it is not used for σ_e (the deflector’s Ca H+K...Mg b absorption falls in the red) but supplies the source-z = 1.302 cross-check (Fe II UV multiplet; §2.2.2). The blue combined cube is not in this repo (only the RL cube) — quote the blue arm from the observing log, not a local file.
- Conditions (Nov-17 frame, header-literal): airmass 1.165 (AIRMASS); guide-star FWHM 1.271" (GUIDFWHM) — the adopted seeing; parallactic angle 15.6°, rotator 0°. (Per-night airmass/seeing for the other two nights not carried in the combined cube.)
- Cube: 100 × 100 spaxels × 3317 λ -pixels; 0.300"/spaxel (square; from WCS), 30" × 30" FOV; wavelength 5625–8941 Å at 1.0 Å/pix (CRVAL3/CD3_3), red central λ 7150 Å (RCWAVE). Spectral resolution $R \approx 10000$ at the fit band → $\sigma_{\text{inst}} \approx 12.6$ km/s (well below $\sigma_e \approx 270$; the LSF is carried into ppxf). [reference_kcwi_data_properties.md]
- Pointing RA/DEC = 02:06:13.38 / −01:14:20.8; target TARGRA/TARGDEC = 02:06:13.58 / −01:14:19.8.

§1.2 HST/WFC3 imaging (F200LP, F140W)

- Program 16773, PI Karl Glazebrook (PROPOSID/PR_INV_*); TARGNAME = DESJ0206-0114; both bands UT 2022-07-14 (EXPSTART MJD 59774.30–59774.31), ORIENTAT 111.9°, drizzled, units ELECTRONS/S.
- F200LP — WFC3/UVIS, aperture UVIS2: EXPTIME 600.0 s, NDRIZIM 4, NCOMBINE 2, drizzle scale 0.050"/pix; PHOTFLAM = 5.1851×10^{−20}, PHOTPLAM = 4923.48 Å → ZP_AB = 27.344.
- F140W — WFC3/IR: EXPTIME 597.7 s, NDRIZIM 3, NCOMBINE 3, drizzle scale 0.080"/pix; PHOTFLAM = 1.4829×10^{−20}, PHOTPLAM = 13922.91 Å → ZP_AB = 26.446.
- AB zeropoints are recomputed per-image from PHOTFLAM/PHOTPLAM (never hardcoded); ZP_AB = −2.5 · log₁₀(PHOTFLAM) − 21.10 − 5 · log₁₀(PHOTPLAM) + 18.6921. Drizzle pixel correlation: empirical noise ≈1.3–2× the CCD-equation value → motivates the 10% flux floor (§2.1.4).

§1.3 JWST/NIRCam imaging (F150W2, F322W2)

- Program 05594 — “JWST Cluster SLICE — Strong Lensing and Cluster Evolution”, PI Guillaume Mahler (SURVEY category); TARGPROP = ACT-CL_J0206.2–0114 (the cluster field; the deflector sits in it). Both bands UT 2024-08-27 (DATE-BEG 06:02–06:40), NIRCam module B, units MJy/sr (i2d), NINTS 1 / NGROUPS 4.
- F150W2 (SHORT channel, CLEAR pupil): EFFEXPTM/XPOSURE 1836.0 s; PIXAR_SR = 2.2222×10^{−14} sr → 0.03075"/pix (PIXAR_A2 = 9.454×10^{−4} ") → ZP_AB = 28.033.
- F322W2 (LONG channel, CLEAR pupil): EFFEXPTM/XPOSURE 1836.0 s; PIXAR_SR = 9.3307×10^{−14} sr → 0.06301"/pix (PIXAR_A2 = 3.970×10^{−3} ") → ZP_AB = 26.475.

- JWST AB from area: $ZP_AB = -6.10 - 2.5 \cdot \log_{10}(\text{PIXAR_SR})$. No PAM correction (drizzled/resampled).
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§2 Methods

§2.1 Photometric Analysis

§2.1.1 Masking

- Use the `F200LP_mask.fits` as the arc mask — NOT the F140W one. [feedback_masking_strategy.md; reference_hst_jwst_data_properties.md; METHODS §II.4]
 - F200LP mask (2512 HST pixels) is hand-tuned to cover the arc only and leaves the deflector core unmasked — it is the original arc-tuned segmentation mask.
 - F140W mask (1121 HST pixels) covers the arc PLUS the deflector core (the photometry tool’s aperture geometry pulled the core in); the two masks have zero spatial overlap.
 - Consequence: F140W is valued for its *image* (sharper Sérsic fit), never for its mask. Using the F140W mask as an arc mask would destroy the deflector-core signal. ### §2.1.1b Principled (reproducible) arc masking — NEW 2026-05-29

The hand-painted masks are now backed by an objective, reproducible recipe (so a referee can reproduce the arc selection), validated + applied to all four bands. [NOTES_arc_mask_verification_2026-05-29.md, NOTES_photometry_mask_systematics_2026-05-29.md; scripts/arc_mask_verification.py, mask_method_comparison.py, photometry_systematics.py]

- Locator = F200LP (highest contrast on the blue $z=1.302$ source). Two independent objective selectors reproduce the F200LP hand mask: (i) a color cut $m_{F200LP} - m_{F140W}$ (arc bluer than the red deflector); (ii) a Sérsic-residual mask (subtract a 2D deflector model, flag pixels whose positive residual exceeds $k \cdot \sigma_{\text{sky}}$, where k is the detection threshold in units of the sky noise σ_{sky} and σ_{sky} is the robust background scatter; we adopt $k = 3$). On F200LP the Sérsic-residual mask reproduces the expert photometry to 0.016 mag, R_e to 3%.
- Transfer to other bands by reprojecting the F200LP footprint (WCS + map_coordinates). Reproduces the expert JWST aperture mags to 0.01–0.02 mag.
- WCS registration (NEW 2026-06-10). A raw WCS reprojection leaves a 0.09–0.18″ residual offset between the F200LP arc and the JWST/F140W frames (HST UVIS↔IR and HST↔JWST astrometry; the JWST i2d GWCS differs from the FITS-header WCS astropy reads). On the thin arc that mis-covers one edge and lets a companion slip past the mask. We cross-correlation-register the reprojected F200LP arc (and the F140W color layer) to each band’s own image near the deflector (photometry_systematics.register_shift), which drives the residual offset to 0.00″ (verified F140W 0.179→0, F150W2 0.154→0, F322W2 0.089→0). Field-galaxy contamination check after registration: no unmasked source ($>8\sigma$) falls inside the photometry aperture in any band. *Do not* anchor on the F200LP deflector centroid instead — it is arc-biased (the faint blue deflector centroid is pulled $\sim 0.26''$ by the arc), so the arc cross-correlation is the right fiducial.
- Deflector model = single-component Sérsic (per user 2026-06-10). The deflector is an elliptical, so a single Sérsic is the physically-motivated light profile — we do not fit a bulge+disk decomposition (an earlier 2-component model was only a numerical kludge to make the residual-based mask growth behave; see the mask-evolution note below). The single Sérsic is fit per band with masked pixels excluded, capturing 66–87% of the box flux (F140W is the weakest, with a localized central residual from the absent bulge/disk split, but that residual sits inside the arc-free core and does not enter the mask).
 - *Implementation gotcha (documented so it isn’t reintroduced)*: astropy.modeling.Sersic2D’s amplitude parameter is $I(r_{\text{eff}})$, not the central peak. Since $I(0)/I(r_{\text{eff}}) = \exp(b_n) \approx 2150$ at $n=4$, initialising amplitude at the data peak over-predicts the centre $\sim 2000\times$ and LevMar collapses the amplitude to ≈ 0 (the whole galaxy then reads as positive residual → catastrophic over-masking).

Initialise amplitude $\approx \text{peak} \cdot \exp(-b_n)$ ($\sim \text{peak} \cdot 0.05$ for $n \approx 3$) and bound it to $(\text{peak} \cdot 1e-3, \text{peak} \cdot 1.5)$, exactly as `seraic_total_photometry.fit_sersic2d`.

- IR extends the source footprint — grown band-specifically (HST color gate / deep-JWST morphology guard): the deeper/sharper JWST bands see the arc’s fuller extent, so we region-grow the F200LP arc seed into the IR. The “is this pixel source, not deflector body?” test is gated two ways depending on the band’s depth (a candidate must also be a $>k \cdot \sigma_{\text{sky}}$ residual, within $R < 5''$, and spatially contiguous with the arc seed — region growing via `scipy.ndimage.label`):
 1. HST bands (F200LP, F140W) — COLOR GATE: the pixel must be bluer than the deflector core by $\geq \Delta C = 0.5$ mag in F200LP–F140W (DCOLOR), at $k = 3$ (K_{EXT}).
 2. Deep JWST bands (F150W2, F322W2) — MORPHOLOGY GUARD: the pixel must be source-dominated, $\text{data} - \text{sky} - \text{model} > \text{MORPH_FRAC} \cdot \text{model}$ (excess beats the single-Sérsic deflector model $\rightarrow$ it is OUTSIDE the Sérsic-dominated region), at the lower $k = 2$ ($K_{\text{EXT_JWST}}$). No HST color is required, so the deep JWST data captures the diffuse arc emission HST is too shallow to color-confirm. Because the single Sérsic already accounts for 66–87% of the deflector flux, $\text{resid} > \text{model}$ rejects the deflector body (model-dominated) while admitting the diffuse source (excess-dominated) — verified: raw mag stable (no deflector loss), 0 unmasked $>8\sigma$ sources in the aperture, the deflector core $<0.4''$ never masked. F150W2 IR-extension grows 1049 $\rightarrow$ 5288 px (the diffuse arc), F322W2 415 $\rightarrow$ 1230 px.
- Why the color gate is needed (and is the right physics). A *single* Sérsic, being the correct elliptical model, still under-fits the bright extended IR galaxy body slightly, leaving a low-level positive residual over the *red* galaxy. Residual-only growth (criterion 1 alone, the old 2-component recipe) therefore sweeps red galaxy-body light — and even neighbouring companions — into the “arc” mask (verified: F150W2 mask balloons and raw mags fall 1–2 mag). The $z=1.302$ source is a blue, star-forming lensed galaxy; the deflector is a red, passive elliptical. Requiring the grown pixels to be bluer than the deflector core (criterion 2) exploits this intrinsic 4000-Å-break contrast to keep the blue source extension while rejecting the red body. The mask thus becomes a property of the source color, not of the (imperfect) deflector residual. This is the same physical contrast used by the validated F200LP color selector (i) above; we are simply applying it per-pixel during the IR growth.
- Color construction. Per target-band grid, reproject the F200LP and F140W surface brightness (`scripts.arc_mask_verification.reproject_intensity`, WCS + bilinear map_coordinates) and form $-2.5 \cdot \log_{10}(\text{SB}_{\text{F200LP}}/\text{SB}_{\text{F140W}})$ (arc $\rightarrow$ more negative). Constant ZP offsets cancel because we threshold on the contrast vs the deflector-core color (median in a 0.4–0.8'' annulus), not on an absolute color.
- Caveats. (a) (*RESOLVED 2026-06-10*) the HST color gate is conservative where HST is shallow and was missing faint diffuse JWST arc emission (user catch via the deep-stretch `_diffuse_mask_check`); the deep-JWST morphology guard above now captures it using JWST’s own sensitivity + the source-dominance test, without over-masking the deflector (raw mag stable). (b) $\Delta C = 0.5$ mag and $\text{MORPH_FRAC} = 1.0$ are tunables; the resulting masks reproduce the expert hand-mask $M_{\star}$ (11.33), our calibration that they neither over- nor under-mask. (c) The morphology guard assumes the single Sérsic captures the deflector majority — true here (66–87%); if a future target has a worse single-Sérsic fit, re-check that $\text{resid} > \text{model}$ still protects the body.
- raw vs fill-in: raw photutils (masked) discards the deflector light under the arc; the single-Sérsic fill-in (replace masked pixels with the model) recovers it. Under the color-gated masks the fill-in correction is -0.19 to -0.27 mag per band (smaller than the old 2-component $+0.18$ to $+0.96$ mag, because the cleaner masks remove less deflector light). The headline is the empirical raw value with the best (per-band, color-gated) mask; the fill-in is carried as a one-sided $+\text{sys}$ (deflector light the mask removes). PSF effect on the fill stays negligible (`scripts/psf_fill_model.py`: ≤ 0.004 mag $\rightarrow \Delta M_{\star} \ll 0.01$ dex; arc is outside the PSF core).
- per-band vs global mask (union of all per-band masks): masking systematic ± 0.086 dex (10%); the raw $\leftrightarrow$ filled (under-arc) term dominates (± 0.057), the per-band $\leftrightarrow$ global mask-definition term ± 0.028 . See §3.2.

- Mask-evolution note (why we settled on the single-Sérsic + color-gated mask). Three attempts, documented in `results/figures/mask_attempts_comparison.png` + notebook 12 §8:
 - (A) 2-component (bulge+disk) Sérsic + residual-only IR growth — over-masks: residual-only growth grabs red galaxy body + a companion (F150W2 15768 px, F322W2 4258 px) → raw biased low (11.16) and a large +0.31 fill-in. The bulge+disk was never physical for this elliptical; it was a kludge to tame the residual growth.
 - (B) single-Sérsic with a broken fit (amplitude→0) + color gate — DISCARDED artifact: the near-zero model made the residual term meaningless and collapsed the fill-in to ~0.01 mag (apparent 11.34 ± 0.013). Caught via the black model panel in the diagnostic figures.
 - (C) single-Sérsic (fixed init) + color-gated IR growth + WCS registration — FINAL: tighter, arc-focused, correctly-registered masks (F150W2 7698 px, F322W2 1997 px; companion now masked), raw central 11.36 consistent with the validated expert hand-mask 11.33, fill real but smaller (−0.19 to −0.28), masking systematic ± 0.086 (vs ± 0.16 for 2-comp). Physically motivated (single Sérsic) + physically-gated (source color) + astrometrically registered (§2.1.1b).
- Files (in sibling dir `../velocity_dispersion_from_IFU/`): `AGEL020613-011417A_F200LP_WFC3_cutout_L3_mask.fits` (500×500, 25"×25"); the mask was revised 2026-02-28 (`_mask_20260228.fits`); original kept as `_mask_OLD.fits`.
- JWST bands: same arc geometry; the HST F200LP mask is re-projected from the HST WCS onto each JWST frame. [METHODS §II.4]
- For the IFU/ σ_e application (context): F200LP mask reprojected to the 0.30" IFU grid via `scipy.ndimage.map_coordinates(order=0) nearest-neighbor`; 35/154 = 23% of spaxels (28% by I-weight) flagged inside $R_e=2.097''$ (a geometric LOWER bound — the 1.27" seeing spreads the arc much wider; see §2.4.1 step 3 + `scripts/psf_mask_fraction.py`). [feedback_masking_strategy.md]

§2.1.2 Estimating R_e

- Headline $R_e = 2.097'' = 14.76$ kpc at $z = 0.67564$ (best-mask F140W+F200LP CoG). [CLAUDE.md; METHODS §II.3/A3]
- “Best mask throughout” (2026-06-11): the headline R_e is now the best-mask (single-Sérsic + color/morph gate + WCS registration) curve-of-growth = $2.097''$, superseding the expert-mask 2.305". The best masks genuinely remove outer diffuse-arc + interloper light that slightly inflated the expert-mask R_e ; the model-based Sérsic r_{eff} (1.897") and sky-subtracted CoG (1.922") both confirm the inward shift. The HST 2- R_e companion mask is empty (companions live only in deep JWST), so it does not enter the HST R_e — the global color/morph mask alone gives $2.097''$ (self-consistent fixed point). `scripts/re_mask_sensitivity.py`, `scripts/Re_bestmask_reconciliation.py`.
- photutils validation (`scripts/validate_Re_photutils.py`): our azimuthally-averaged masked CoG is independently reproduced by photutils `RadialProfile` (integrated) to $\pm 0.002''$, and unmasked matches photutils `CurveOfGrowth` (direct 2D aperture sum) to $\pm 0.004''$. A naive masked aperture-*sum* (photutils `CurveOfGrowth` dropping masked pixels) would bias $R_e +0.25\text{--}0.41''$ high → our azimuthal-fill is the correct masked-CoG treatment. `centroid_2dg` is robust; `centroid_quadratic` fails on extended light (expected — a point-source tool).
- Method systematic (best mask) = $\pm 0.100''$ (raw CoG 2.097 / sky-sub CoG 1.922 / Sérsic r_{eff} 1.897, peak-to-peak/2; `results/Re_cog_reconciliation_bestmask.npz`).
- Method: mean of the F140W and F200LP best-mask curves-of-growth (`scripts/final_sigma_e.py:curve_of_growth`, the PRODUCTION script):
 - best-mask F140W CoG = 1.912"; F200LP CoG = 2.281"; mean = $2.097''$ (expert-mask values were 2.168/2.441/2.305).
 - Masked CoG zeros mask-flagged HST pixels before the radial flux integral, so the lensed arc / companions do not bias the half-light point.
 - Centers: HST-mean `photutils.centroid_2dg` centers (same as the σ_e pipeline), both bands.
- Do NOT confuse with `scripts/measure_Re.py` → `results/Re_measurements.npz`. That script uses the image-geometric center and produces 10 variants (3 sources × 4 masking strategies: unmasked, zeroed,

proper, PSF-convolved); its “proper-mask” mean (2.633”) is not the headline — early-reference only. (measure_Re.hst_Re previously hardcoded pixscale=0.08 for both bands; fixed 2026-06-08 (A3c/nb14) to read the WCS pixscale + r_max_arcsec=6.0, now reconciling with the production CoG to <0.04” — but final_sigma_e remains the headline path.)

- Superseded prior values: IFU white-light R_e = 2.61” (nb05/06 era); expert-mask HST CoG = 2.305” (pre-2026-06-11). A Ca H+K+G depth-map gives 2.90” (kinematic-tracer estimator, not the photometric R_e).
- R_e as a σ_e systematic (test D7 → best-mask grid): σ_e rises monotonically with R_e. Re-measured at the wide window over a 7-point R_e grid bracketing the 2.097” headline (scripts/run_sigma_e_Re_grid.py); the adopted R_e-source systematic = ± 5.13 km/s (best-mask CoG family {1.912/2.097/2.281”}), folded into the budget (§3.1, §2.4.3). The full grid incl. CaHK+G 2.90” (± 9.98) and CaHK+G’s +12.4 km/s deviation are cross-checks, NOT folded. [CLAUDE.md]

§2.1.3 Measuring Fluxes

- Tools: photutils aperture photometry (interactive masking GUIs scripts/photometry_masking_{HST,JWST}.py). **synphot** is NOT currently used — flux→AB is done directly from header zeropoints. (*If the outline wants synphot, that is aspirational; flag with co-authors.*) [METHODS §II.4]
- AB zeropoints — read per-image from FITS headers, never hardcoded. [CLAUDE.md; reference_hst_jwst_data_properties.m]
 - HST (drizzled e-/s): ZP_AB = $-2.5 \cdot \log_{10}(\text{PHOTFLAM}) - 21.10 - 5 \cdot \log_{10}(\text{PHOTPLAM}) + 18.6921$.
 - JWST (i2d, MJy/sr): ZP_AB = $-6.10 - 2.5 \cdot \log_{10}(\text{PIXAR_SR})$.
- Four bands / instruments / programs:

Telescope	Instrument	Filter	Program (PI)	Drizzled scale	EXPTIME	ZP_AB	AB (aperture)
HST	WFC3/UVIS	F200LP	16773 (Glazebrook)	0.0500"/pix	600.0 s	27.344	22.613
HST	WFC3/IR	F140W	16773 (Glazebrook)	0.0800"/pix	597.7 s	26.446	19.134
JWST	NIRCam SW	F150W2	05594 (Mahler)	0.03075"/pix	1836.0 s	28.033	18.942
JWST	NIRCam LW	F322W2	05594 (Mahler)	0.06301"/pix	1836.0 s	26.475	18.604

[reference_hst_jwst_data_properties.md] - JWST pointing comes from the cluster program ACT-CL J0206.2–0114 (ACT DR5, Hilton et al. 2021); the deflector sits in that field. No PAM correction (drizzled). Drizzle pixel correlation: empirical noise $\sim 1.3\text{--}2\times$ the idealized CCD-equation value — motivates the 10% flux floor (§2.1.4). - 2D Sérsic verification given masking (the morphology cross-check; notebook 08): single-component 2D Sérsic per band (scripts/measure_Re.py framework + the 2026-05-11 bound-fix $n \in [1.0, 6.0]$, $\text{ellip} \in [0.0, 0.6]$, 3-init grid — prevented an $n=0.30$ flat-disk escape on F200LP), extrapolated to total flux:

Filter	AB aperture	AB Sérsic-total	Δmag	n	r_{eff} (")
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Aperture under-counts outer-profile flux by 0.8–1.9 mag, but the integrated-mass effect is only +0.07 dex (see §2.1.4 / Results). Caches: results/sersic_total_photometry.npz, results/bagpipes_sersic_refit.npz. [METHODS §II.6]

§2.1.4 SED Modeling with bagpipes

- Tool: bagpipes (Carnall et al. 2018, 2019); BPASS+CLOUDY templates (Bagpipes default). Notebook 02_streamlined_Bagpipes_SED.ipynb. [METHODS §II.5; reference_sed_fitting.md]

- Model: exponential- τ SFH; Calzetti dust (A_V free, 0–2); free metallicity (0–2.5 $Z_\odot$); redshift prior (0.674, 0.676) bracketing the line-fit z . Sampler Nautilus ($n_{\text{live}}=400$), 500-sample posterior.

```
fit_instructions = {
  "redshift": (0.674, 0.676),
  "exponential": {"age": (0.1, 15.), "tau": (0.3, 10.),
                  "massformed": (1., 15.), "metallicity": (0., 2.5)},
  "dust": {"type": "Calzetti", "Av": (0., 2.)},
}
```

- Why a 10% fractional error floor on each band: a conservative, uniform systematic budget that covers both the formal photon noise and the drizzle pixel-correlation underestimate (empirical noise $\sim 1.3\text{--}2\times$ the idealized CCD-equation value). [METHODS §II.4/II.7; reference_sed_fitting.md]
- Systematic test to fill in masked pixels = the 2D-Sérsic-total refit: because the aperture excludes arc-masked pixels and truncates the outer profile, we fit a 2D Sérsic per band (§2.1.3), extrapolate to total flux, and re-run Bagpipes with identical priors. This recovers the flux missing from masked/truncated regions and bounds the mass bias. Result: $\log(M_\star/M_\odot) = 11.40 \pm 0.11 \text{--} 0.15$ (Sérsic-total) vs $11.33 \pm 0.07 \text{--} 0.09$ (aperture) $\rightarrow$ aperture is biased low by ~ 0.07 dex (+16% in $M_\star$), within the quoted uncertainty. [METHODS §II.6] NOTE (updated 2026-06-10): this analytic-total refit is superseded by the per-aperture single-Sérsic fill-in under the principled color-gated IR-extended masks (§2.1.1b / §3.2): the headline is the empirical raw value 11.36 (10%) with a one-sided +0.14 dex fill-in systematic reaching 11.47. The 11.40 analytic-total sits just inside that bracket and the central is consistent with the validated expert hand-mask 11.33. The enhanced comparison figure `results/figures/nb08_sersic_vs_aperture.png` now overlays, for both the empirical-aperture and Sérsic-total choices, the best-fit Bagpipes model SED, the measured photometry, and the filter-convolved model photometry.
- Posteriors (older expert-aperture, now a cross-check): $\log(M_\star/M_\odot)=11.33 \pm 0.07 \text{--} 0.09$; mass-weighted age 5.10 Gyr [3.69–6.11]; $\tau = 0.68$ Gyr; $Z = 0.89 Z_\odot$; $A_V = 0.82$ mag; $z_{\text{phot}} = 0.675$ [0.674–0.676]. Per-band residuals: F200LP +4.7%, F140W -11.3% , F150W2 -8.0% , F322W2 +9.6% (mild single- τ -SFH NIR-slope tension; not catastrophic, within $\pm 0.07\text{--}0.15$ dex). [METHODS §II.5]
- Caches / figure: `results/bagpipes_sed_results.npz` (the file Figure 2 loads), `pipes/posterior/AGEL0206/0206_real.h5` (delete to refit), `results/bagpipes_sersic_refit.npz`.

§2.2 Spectroscopic Analysis

§2.2.1 Calibration — @Keerthi / Kaustubh

- Instrument: Keck II / KCWI, KCRM red arm, Red-Low (RL) grating, Medium slicer, 2×2 binning; dithered at PA $0^\circ/45^\circ/90^\circ$. [METHODS §I.1; reference_kcwi_data_properties.md]
- Cube (headline): Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits — the NEW `_mtwdo_` reduction (Master Twilight + Dome hybrid flats), K. R. Gupta’s latest re-reduction. The earlier `no_mtwdo_` cube is the OLD reduction (second-reduction cross-check; co-aligned to $<0.002''$, differs only in flat-fielding; σ_e differs ~ 7 km/s — the “reduction-pass” systematic).
- Reduction chain: raw frames $\rightarrow$ `kcwidrp`; stacked with `ISMgas/kcwi/kcwiReduction` by K. R. Gupta (Kaustubh); Dec-29 RED frames pre-reduced by Yuguang Chen. (*Calibration prose is Keerthi/Kaustubh’s to write — these are the facts on hand.*)
- Cube geometry: shape (3317, 100, 100); $0.300''$ spaxels; $\sim 30''\times 30''$ FoV; $\lambda = 5625\text{--}8941 \text{ \AA}$ at $\Delta\lambda = 1.0 \text{ \AA/pix}$; BUNIT = FLAM16/arcsec².
- Three confirmed nights (totals ≈ 180 min RED + 198 min BLUE on source):

Night (UT)	Program (PI)	RED	BLUE
2024 Dec 29	U204	4 $\times$ 300 s	1 $\times$ 1320 s (excluded by reducer)
2025 Aug 30	K409 (PI TBD)	12 $\times$ 300 s	4 $\times$ 990 s

Night (UT)	Program (PI)	RED	BLUE
2025 Nov 17	U002 (Jones)	20×300 s	5×1320 s

FLAG: the cube header (DATE-OBS 2025-11-17, PROGPI Jones) describes only the first input frame — cite the multi-night combine, not the header. K409 PI and Aug-30 DIMM seeing still TBD. - Spectral resolution / LSF: $\text{FWHM}_{\text{inst}} = 2.355 \times \text{DISPSCAL}(0.2940) \times 1.0 \text{ \AA} = 0.692 \text{ \AA} \rightarrow R \approx 10000$ at the fit band; $\sigma_{v,\text{inst}} \approx 12.6 \text{ km/s}$ observed (7.5 km/s rest-frame) — the $\sim 270 \text{ km/s}$ dispersion is resolved by $\sim 20\times$. LSF robustness $\leq 0.83 \text{ km/s}$ over $\text{DISPSCAL} \times 0.5\text{--}2.0$. - Seeing: guide-camera FWHM = $1.27''$ (GUIDFWHM, Nov-17; conservative).

§2.2.2 Redshift Measurement

- Method — single/multi-line Gaussian fits to the integrated-spectrum features on NIST air rest wavelengths, per-line z inverse-variance combined (`scripts/redshift_verify.py`, notebook 04).

Deflector absorption features (systemic $z = 0.67564 \pm 0.00033$; 6 fits):

Feature	Ion/species	rest λ_{air} (Å)	obs λ_{air} (Å)	z_{line}	σ (Å)
Ca K	Ca II	3933.66	6591.79	0.67574 ± 0.00096	11.5
Ca H	Ca II	3968.47	6649.61	0.67561 ± 0.00097	16.6
H δ	H I	4101.74	6871.80	0.67534 ± 0.00058	3.3
G-band	CH (band)	4304.40	7209.31	0.67487 ± 0.00160	11.3
H β	H I	4861.33	8149.30	0.67635 ± 0.00082	2.8
Ca H+K	Ca II doublet	3933.66 / 3968.47	—	0.67567 ± 0.00069	11.2
Weighted				0.67564 ± 0.00033	

Source emission features (lensed arc, $z = 1.30263 \pm 0.00003$; [O II] doublet, red cube):

Feature	rest λ_{air} (Å)	obs λ_{air} (Å)	z_{line}
[O II] 3726	3726.03	8579.2	1.30357 ± 0.00005
[O II] 3729	3728.82	8585.6	1.30185 ± 0.00005
[O II] doublet			1.30251 ± 0.00004

- Deflector systemic $z = 0.67564$ (supersedes AGEL DR2 $z = 0.67511$; $\sim 95 \text{ km/s}$ offset, used only in old notebooks). [METHODS §I.2; reference_redshift_verification.md]
- Other absorption features in ABSORPTION_LINES (H γ 4340.47, Fe4383 4383.55, Mg I/Mg b 5175.27, Fe I 5270.40, Na D 5893.00) are NIST-air but were NOT used in the systemic fit (out of the red-cube range at $z=0.676$, or blended); listed for completeness in `scripts/redshift_verify.py`.

Air $\leftrightarrow$ vacuum audit (2026-06-11 — PASS). (1) Every rest wavelength is the NIST air value, verified to $<0.01 \text{ \AA}$ (Ca K 3933.66 vs NIST 3933.663, H β 4861.33 vs 4861.333, [O II] 3726.03 vs 3726.032, etc.). G-band 4304.40 is the CH molecular bandhead (a blend centroid, not a single atomic line) $\rightarrow$ it is the most discrepant/uncertain line ($z=0.67487$, ± 0.0016) and is down-weighted by its error. (2) The KCWI cube is vacuum (FITS CTYPE3= 'WAVE'; AWAV would denote air) and is converted to air before fitting via the Ciddor (1996)/Morton (2000) IAU-standard index of refraction $n = 1 + 8.34254 \times 10^{-5} + 2.406147 \times 10^{-2}/(130-s^2) + 1.5998 \times 10^{-4}/(38.9-s^2)$, $s = 10^4/\lambda_{\text{vac}}$ — the same formula the `sigma_e ppxf` pipeline uses (`bootstrap_ppxf.py`; Ciddor 1996). (3) Thus $z = \lambda_{\text{obs,air}}/\lambda_{\text{rest,air}} - 1$ combines both wavelengths in the air frame $\rightarrow$ internally consistent, z unbiased. Had we left the cube in vacuum and used air rest lines, z would be biased high by $\approx (1+z) \cdot (n-1) \approx +5 \times 10^{-4}$ ($\approx +90 \text{ km/s}$); the conversion removes this. - Double verification with `ppxf`: the Gaussian line-fit z is independent of, and consistent with, the `ppxf V_sys` (`ppxf` returns z via $z = (1+z_0) \cdot \exp(V/c) - 1$; Cappellari 2023, the `ppxf` relativistic velocity convention — *eq. number not yet verified, do not assert “eq. 5c”*). - Source $z = 1.30263 \pm 0.00003$ — measured from the [O II] $\lambda\lambda 3726, 3729$ doublet in the red cube (notebook 04;

obs 8579.2/8585.6 Å), consistent with AGEL DR2 $z = 1.302$. (The ± 0.00003 is the formal doublet-fit error; the per-line spread 1.30185–1.30357 is the realistic uncertainty.) - Source rest-UV resonance lines (arc ISM; `scripts/source_uv_redshift.py`, 2026-06-11). The bright star-forming arc shows the classic Mg II / Fe II resonance absorption: - Mg II $\lambda\lambda 2796.35, 2803.53$ (red arm): strongly detected ($16.7\sigma / 18.4\sigma$), $z(\text{absorption}) = 1.30113 \pm 0.00003$, i.e. a -195 km/s centroid offset vs the systemic [O II] $z=1.30263$ (both in the *same* red cube, so it is a real *relative* offset, not cross-arm calibration). We do NOT interpret this as an outflow without further work — for a resonance line on a *lensed* arc the offset is degenerate among: (i) Mg II emission infill filling the red absorption wing (P-Cygni; pulls the centroid blueward with no bulk motion), (ii) differential lensing / aperture — the Mg II absorption (continuum-weighted) and [O II] emission (H II-region-weighted) sample different source regions that the lens positions/magnifies differently, and (iii) the blended-doublet single-Gaussian centroid on an asymmetric profile. Report it as an observed Mg II–[O II] velocity offset, with outflow as one (unconfirmed) possibility. - Fe II $\lambda\lambda 2344.21, 2382.77$ (blue arm): detected ($7.9\sigma / 3.0\sigma$; Fe II 2374 only 1.2σ), but at $z \approx 1.306$, $\sim +470$ km/s from the red systemic. Because Fe II sits in the separate blue cube (an older AWAV/air reduction, $\neq$ the red WAVE/vacuum mtwdo cube), this offset is dominated by a blue $\leftrightarrow$ red wavelength-calibration difference, so the Fe II *velocity* is unreliable cross-arm — the detection is real, the velocity is a [TODO: cross-calibrate the blue arm]. (Rest λ are Morton 2003 vacuum; the blue cube is converted air $\rightarrow$ vac for the fit. NOT a clean independent z — quote Mg II / [O II].) - Companion ellipticals — a galaxy GROUP at the deflector redshift (notebook 04a, `scripts/redshift_verify.bootstrap_redshift`, $N=200$ wild bootstrap on the red cube). Two early-type companions flank the primary deflector, both at $z \approx 0.676 =$ the deflector z (physically associated, not interlopers):

Companion	position	sep. from deflector	z (absorption)	z (emission)
NE (“2 o’clock”, larger)	IFU (x,y)=(61,64)	$\approx 4.5'' \approx 31$ kpc NE	0.67580 –0.00067/+0.00062	0.67565 –0.00045/+0.00060
SW (“7 o’clock”, compact)	IFU (x,y)=(38,45)	$\approx 4.6'' \approx 32$ kpc SW	0.67588 –0.00034/+0.00047	0.67558 –0.00048/+0.00071

Both companions sit within $|\Delta cz| \lesssim 60$ km/s of the deflector (0.67564) $\rightarrow$ AGEL J020613–011417 is the central galaxy of a compact group/triple at $z \approx 0.676$. (These are the field ellipticals masked in the aperture photometry; their group membership confirms they are not background interlopers.) - Peculiar velocities: we do not correct for peculiar velocities — we use the line-fit systemic z directly (see gap G3; this sentence needs to be added to the draft). - Wavelength convention: AIR rest wavelengths throughout (NIST-verified to <0.01 Å); KCWI native vacuum (CTYPE3= 'WAVE') converted to air via Ciddor 1996 — see the air $\leftrightarrow$ vacuum audit box above. Do not mix SDSS vacuum values (~ 1 Å ≈ 80 km/s systematic at 4000 Å). [feedback_wavelength_convention.md]

§2.2.3 Stellar Kinematics & Velocity Dispersion

Spatial regime / masking / R_e

- Headline σ_e is measured on a single $I(r)$ -weighted 1-D aperture spectrum at $R < R_e$ — the co-add-then-fit method of Cappellari et al. 2006 (SAURON IV, §2.3), the literature-standard σ_e (KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA convention). $R_e = 2.097'' = 14.76$ kpc (best mask; §2.1.2).
- Center: HST-mean photutils.centroid_2dg of F140W+F200LP cores, propagated through the KCWI WCS; adopted RA = 31.55613°, Dec = -1.23819° (ICRS). F140W $\leftrightarrow$ F200LP offset 0.36" (F200LP shifted by the UV-bright arc; F140W is the clean bulge centroid).
- Spatial arc mask: F200LP segmentation mask reprojected to the IFU grid (nearest-neighbor); 38/184 spaxels inside R_e flagged ($\sim 27\%$ by I-weight). F140W mask not used (covers the core).
- I-weight map: per-spaxel collapse of the IFU 6500–7500 Å white-light band; spectra summed (not averaged) so the wild bootstrap preserves per-spaxel noise.
- $R < R_e/8$ (JFK95 σ_c) was dropped — at $0.288''$ it sits inside seeing $\text{FWHM}/2 = 0.64''$ (~ 3 spaxels). No central σ is quoted; $\sigma_e(<R_e/2) \approx 225$ km/s is reported as a gradient diagnostic only.

ppxf setup

- What ppxf does — Penalized Pixel-Fitting (Cappellari & Emsellem 2004, PASP 116, 138; upgraded in Cappellari 2017, MNRAS 466, 798; Cappellari 2023 — *volume/page not re-verified here*). It forward-models the observed galaxy spectrum, in pixel space on a log-rebinned (constant km/s per pixel) grid, as a non-negative linear combination of stellar templates convolved with a parametrised line-of-sight velocity distribution (LOSVD), plus low-order additive Legendre polynomials that absorb continuum/template mismatch. The template weights are solved by non-negative least-squares and the LOSVD parameters by Levenberg–Marquardt, alternately, to minimise χ^2 . The “*penalized*” is an optional bias that pulls the higher Gauss–Hermite moments toward a Gaussian at low S/N — inactive here, since we fit a pure Gaussian LOSVD.
- ppxf v9.4.5, `moments = 2` (fit only V and σ ; no h_3/h_4), bias left at the default (irrelevant at `moments=2`), `clean = False` (cosmic rays are handled by the `BAD_PIXELS_REST` mask instead — ppxf’s internal σ -clip rejects 0 pixels here because the KCWI noise array is over-estimated vs the residual scatter), `velscale` set to the cube’s native sampling, `trig = False`. [`scripts/bootstrap_ppxf.py`, `run_wide_sigma_e.py`; *METHODS §I.5/I.6*]
- Polynomial degrees: additive Legendre degree swept over 15–29 (15 values) and pooled; `mdegree = 0`. σ -vs-degree is flat within the bootstrap envelope (no LOSVD absorption by the polynomial). (*Note: METHODS §I.5 prose mis-labels these “multiplicative” — the code passes `degree=...`, `mdegree=0`, i.e. additive. Describe them as additive.*)
- Three SPS libraries, pooled at the posterior level: FSPS, EMILES, XSL (8–9 templates each over rest 3500–5000 Å).
- Per-SPS native wavelength frame: FSPS = vacuum, EMILES = air, XSL = air. Galaxy frame matched per-SPS (scalar-median $\text{vac} \leftrightarrow \text{air}$ via Ciddor 1996, following Cappellari’s pattern). Why: the legacy “convert galaxy to air for all three” produced a spurious -90 km/s FSPS V_{sys} ; the fix collapsed the V_{sys} split-track $\sim 110 \rightarrow \sim 15$ – 18 km/s (σ shift ≤ 2 km/s). [`reference_ppxf_vacair_handling.md`]
- Per-SPS V_{sys} subtracted before pooling (σ is V-invariant; a single global V_{sys} would inflate σ_e by ~ 10 – 15 km/s in the discrete cross-check). All three libraries now converge to the same redshift — at the headline $R_e=2.097''$ aperture the per-SPS V_{sys} are FSPS -15.7 / EMILES -19.1 / XSL -14.9 km/s $\rightarrow z = 0.67555$ / 0.67553 / 0.67556 (a ~ 4 km/s, $\Delta z \approx 3 \times 10^{-5}$ spread; mean $z_{\text{ppxf}} = 0.67555 \pm 0.00001$). This convergence is *the* signature that the $\text{air} \leftrightarrow \text{vacuum}$ frame fix worked: the legacy “convert all three to air” left FSPS (the vacuum-native library) ~ 90 – 110 km/s discrepant from EMILES/XSL; per-SPS native framing collapses it to the ~ 4 km/s seen here (residual folded into the ± 5 km/s frame systematic).
- Emission-line handling (sigma clipping / removing emission lines):
 - Custom no-Balmer goodpixels (`_determine_goodpixels_no_balmer`): masks only the forbidden lines ([O II], [O III], [O I], [N II], [S II]) and keeps H δ , H γ , H β IN the fit (they are absorption in this passive deflector; masking them discards ~ 120 stellar pixels). H δ : keep unmasked — RESOLVED (M12/nb16): H δ is well-fit (local-MAD not an outlier), and masking it shifts σ_e by $+6$ – 8 km/s = LOSVD *information*, not contamination (the M9 “it’s signal” pattern); the `bootstrap_ppxf.py` TODO is closed.
 - $z = 1.302$ source-emission masks (`ARC_MASKS_REST`, mapped by $(1+z_s)/(1+z_l)=1.374$): Mg II 2796/2803 (def-rest 3835–3855 Å), [O II] 3727/3729 (5115–5135), [Ne III] 3869 (5260–5340), He I 3819 (5237–5253, added 2026-05-27 as test M8).
 - Tellurics — handled by masking, not correction. One band is masked: the O $_2$ A-band leading-edge residual at obs 7593–7626 Å = def-rest 4525–4545 Å (`ARC_MASKS_REST`). Confirmed telluric, not $z=1.302$ source, on 2026-05-18 by the same-observed- λ test (the spike sits at the *same obs* λ in the deflector aperture, the sky box, and a 4–8" off-deflector spectrum). No full telluric correction is applied; the M10 OH/sky bands (below) catch sky-emission residuals.
 - M10 sky-line audit: bad-pixel catalog `BAD_PIXELS_REST` = 35 entries (26 CR residuals + 9 OH air-glow/sky bands), all verified non-source via the arc spectrum.
- Wild bootstrap errors ($\rightarrow$ paper §3.2.4; full detail in §2.2.3.4 below). Hybrid Rademacher sign-flip $\times$ local-residual scaling, $N = 500$ production per (SPS $\times$ degree), errors = 16/84 percentiles. See §2.2.3.4 for the exact algorithm and the no- σ -clip note.

§2.2.3.4 Bootstrap error analysis (paper §3.2.4)

This is the σ_e statistical error. It is computed by a hybrid wild bootstrap on the I-weighted $R < R_e$ aperture spectrum, run independently for every (SPS library $\times$ additive polynomial degree) and then pooled. [scripts/bootstrap_ppxf.py: compute_local_residual_scaling, run_bootstrap_single_degree, run_bootstrap]

- The algorithm, per (SPS $\times$ degree):
 1. Residuals $r = \text{galaxy} - \text{best-fit}$, where best-fit is the original (un-resampled) ppxf fit at that degree.
 2. Heteroskedastic local rescale (the “local-residual” half). A 75-pixel rolling window (`uniform_filter1d, mode='reflect'`) gives the local residual std at each pixel; the scale factor = $\text{local_std} / \text{global_std}$, clipped to $[0.2, 5.0]$. $\text{scaled_r} = r \times \text{scale}$. This preserves the wavelength-dependent noise structure (sharp line cores vs smooth continuum) that a flat resample would erase.
 3. Rademacher draw (the “wild” half). Each pixel gets an independent sign $s \in \{-1, +1\}$ with equal probability; the synthetic spectrum is $\text{galaxy_boot} = \text{best-fit} + \text{scaled_r} \times s$. The sign-flip keeps the pixel grid, the LOSVD line shape, and the local noise amplitude intact — you cannot resample pixels independently here without destroying the line profile that σ is measured from.
 4. Re-fit ppxf on galaxy_boot with the *same* goodpixels, `moments=2`, `degree`, `mdegree=0`; record V , σ , χ^2 , and $z = (1+z_0) \cdot \exp(V/c) - 1$.
- $N = 500$ production per (SPS $\times$ degree); $N = 50$ smoke agrees to within 1 km/s. RNG seeded (`np.random.default_rng(seed)`) for reproducibility. Errors = 16th/84th percentiles (asymmetric).
- Pooling = the systematic-marginalisation step. All three SPS (FSPS+EMILES+XSL) $\times$ 15 polynomial degrees (15–29) bootstrap samples are pooled into one posterior. The pooled width therefore already marginalises over the SPS-library and polynomial-degree systematics — between-SPS spread ± 2.04 , within-SPS bootstrap ± 4.22 , quadrature $4.73 \approx$ pooled 4.64 $\rightarrow$ no separate SPS or degree line is added to the budget (§3.1).
- Headline stat error: ± 4.6 km/s (asymmetric $-5.16 / +4.13$) at the $R_e = 2.097''$ aperture.
- No in-fit σ -clipping — `clean=False` (the ppxf default; `clean=` is not passed, `bootstrap_ppxf.py:434`). Rationale: `clean=True` rejects 0 pixels here because the KCWI noise array is over-estimated relative to the actual residual scatter, so iterative σ -clipping is a no-op. Outliers are instead removed up front by the explicit 35-entry `BAD_PIXELS_REST` mask (26 CR residuals + 9 OH/sky bands, all verified non-source via the arc spectrum), so every bootstrap iteration fits the same fixed `goodpixels` set — no per-iteration rejection that could bias σ .
- How the bad pixels were flagged (the reportable σ -clip number). `BAD_PIXELS_REST` was built by a rolling 75-pixel local-MAD on the canonical ppxf-fit residuals, clipping at $|\text{residual}|/\text{local-MAD} > 3\sigma$ (rest 3800–5400 Å) $\rightarrow$ 52 outlier pixels in 26 contiguous ranges (~ 46 Å, padded ± 0.5 Å to survive log-rebinning); biggest outlier = a 6-pixel cluster at def-rest 5232–5236 Å (obs 8768–8774 Å) at 26σ in local-MAD units. Intrinsic to the data (same pixels in both `_mtwdo_` and original reductions). The M10 sky-line audit then added 9 OH/airglow bands by a separate threshold — $> 2.5\times$ the median sky std — all arc-verified non-source. [scripts/run_window_sweep.py:112–155] Report as: “cosmic-ray residuals flagged at 3σ in a 75-pixel rolling local-MAD; sky/OH bands added above $2.5\times$ the median sky noise (35-entry catalog) — distinct from the in-fit `clean=False`.”
- Performance: `joblib` parallel, BLAS pinned to 1 thread/worker (avoids oversubscription); ~ 12 min per SPS library at $N=500$. Outputs: `results/ppxf_bootstrap_errors_{sps}_{_suffix}.npz`, arrays shape (`n_degree`, 500) for $V/\sigma/\chi^2/z$ plus p16/p50/p84 summaries and asymmetric errors.
- Citation TODO (honest): the repo cites no wild-bootstrap reference. The standard statistics precedents for the Rademacher-weight wild bootstrap are Wu (1986), Mammen (1993), Davidson & Flachaire (2008) — *not yet in Zotero; verify + add before drafting, do not assert a specific one as the method source until checked.*

σ_e definition & wavelength window

- σ_e definition — the luminosity-weighted second moment of the LOSVD within R_e . (Defined as the radial integral in Gültekin et al. 2009, eq. (1) [verified 2026-06-12]; the relation we place it on is Kormendy & Ho 2013 eq. 3.) Two mathematically-equivalent representations:
 - Analytic / luminosity-weighted integral — Gültekin et al. 2009 eq. (1) (their 1D form; generalised to the 2D axisymmetric area element $dA = 2\pi r dr$):

$$\sigma_e^2 = \frac{\int_0^{R_e} (V^2 + \sigma^2) I(r) 2\pi r dr}{\int_0^{R_e} I(r) 2\pi r dr}.$$

- Discrete spaxel/bin sum — the same moment evaluated over the N spaxels (or Voronoi bins) i inside R_e , with F_i = the I-band flux (luminosity weight) of spaxel i ; the $2\pi r$ factor is captured implicitly because the number of spaxels per annulus scales with area:

$$\sigma_e^2 = \frac{\sum_{i=1}^N F_i [(V_i - V_{\text{sys}})^2 + \sigma_i^2]}{\sum_{i=1}^N F_i}.$$

This discrete spaxel/bin sum is Cappellari et al. 2013 (ATLAS3D XV, MNRAS 432, 1709) eq. 29 — $\langle v_{\text{rms}}^2 \rangle_e \equiv \langle V^2 + \sigma^2 \rangle_e \equiv \sum_i F_i (V_i^2 + \sigma_i^2) / \sum_i F_i$ — equivalently the discretised form of the Gültekin 2009 eq. 1 integral. (Their eq. 29 also carries an inclination/projection element in the dynamical-model framework; our §7 uses the *observed-plane* projected V_n, σ_n directly and omits that term.) It is NOT “Cappellari 2006 eq. 1” — a prior draft mis-cited it there; Cappellari 2006 eq. 1 is the aperture correction $(\sigma_R/\sigma_e) = (R/R_e)^{-0.066}$, and its σ_e (SAURON IV §2.3) is the co-add-then-fit method below. [eq. numbers verified 2026-06-12: Cappellari 2006 eq. 1, Cappellari 2013 eq. 29, Gültekin 2009 eq. 1]

- Both forms are implemented and cross-checked here (do not re-derive). The headline (§6cum/nb09) is the co-add-then-fit method of Cappellari et al. 2006 (SAURON IV, §2.3) — co-add the I-weighted $R < R_e$ spectra into one effective spectrum and fit a single LOSVD with pPXF; the width of that co-added spectrum *equals* the luminosity-weighted second moment above (this is what KH13/Greene+20/ATLAS^{3D} report). The discrete annular sum (Gültekin 2009 eq. 1) is the §7 / nb07c cross-check (run_gulteкин_mc, F_j = ΣI over each unmasked annulus). The two agree at $<1\sigma$: co-add-fit (§6cum, narrow) 267.32 ± 24 vs discrete annular (§7, arc-filtered to $R < R_{\text{safe}} = 3R_e/4$) $256.17 - 13.0/+12.7$. Full implementation + subtleties in reference_gulteкин_implementation.md.
- ★ What the headline code ACTUALLY computes (final_sigma_e.extract_aperture_spectrum → pPXF; this is the number we report, 267.31). We do NOT evaluate either σ_e^2 sum or integral above. We form the $I(r)$ -weighted co-added aperture spectrum

$$S_e(\lambda) = \sum_{i=1}^N w_i S_i(\lambda), \quad w_i = \frac{I_i}{I_{\text{tot}}}, \quad I_{\text{tot}} \equiv \sum_{i=1}^N I_i, \quad I_i \equiv 0 \text{ for arc-masked spaxels (mask_weight=0),}$$

where i indexes the N spaxels inside R_e ($r_i < R_e$), $S_i(\lambda)$ is the spaxel spectrum, and I_i its 6500–7500 Å observed-band IFU flux (the luminosity weight; arc-masked spaxels carry $I_i = 0$ so $w_i = 0$). The denominator I_{tot} is the single normalisation constant (total in-aperture flux) so $\sum_i w_i = 1$. The choice of weight map I_i is itself a budgeted systematic — the “I-shape” term (± 2.29 km/s; §2.4.3, run_isource_shape_sweep.py + ishape_arcfree_pooled.py). Holding the spaxel selection fixed, we re-measure σ_e with 14 alternative weight maps: the 10 of run_isource_shape_sweep — (1) the headline 6500–7500 Å IFU band,

- (2) full IFU white-light, (3–4) HST F140W / F200LP reprojected (raw), (5–6) the same arc-masked, (7–8) their 1-D curve-of-growth annular means, (9–10) their 2-D Sérsic-model fits — plus 4 arc-free PSF-matched maps (11–14: F140W/F200LP single-Sérsic and arc-filled, each convolved to the 1.27″ KCWI PSF; see below). Peak-to-peak/2 of the full 14-shape spread (266.79–271.37 km/s, NEW cube + M10) = ± 2.29 km/s. So I_i = IFU white-light is the *headline* choice, not an assumption: σ_e moves $\leq \pm 2.29$ km/s across this whole family of luminosity weightings.

- PSF resolution of the HST weight maps (FOLDED into I-shape 2026-06-12). The HST F140W/F200LP (and Sérsic) weight maps are reprojected at native HST resolution, not convolved to the KCWI $\sim 1.27''$ seeing. To test resolution-matching correctly the arc must be removed first — using the *arc-free deflector model* (single-Sérsic, or the HST image with the masked arc filled by that model) and only *then* convolving to the KCWI PSF (`scripts/ishape_arcfree_psf_test.py`, `ishape_arcfree_pooled.py`). On the 3-SPS-pooled footing these arc-free, PSF-matched weights give $\sigma_e = 266.79\text{--}268.78$ km/s (F140W Sérsic 266.79 / filled 267.38; F200LP Sérsic 268.29 / filled 268.78) — i.e. they fall inside the existing 10-shape spread (266.83–271.37). Folding them in (14 shapes) moves the I-shape term $\pm 2.27 \rightarrow \pm 2.29$ (negligible) and σ_e is unchanged. (*Cautionary cross-check: naively convolving the raw arc-containing HST images instead gives a spurious $+7\text{--}8$ km/s — that is arc leakage (the $1.27''$ kernel spreads the masked arc light into unmasked arc-adjacent spaxels, worst for the UV-bright F200LP) plus high- σ -outskirt up-weighting, NOT a resolution effect; not used.*) The headline IFU white-light weight is already at KCWI resolution AND arc-masked AND deflector-light-weighted, so the resolution choice is bounded within the I-shape term. pPXF then fits a single Gaussian LOSVD (moments=2) to $S_e(\lambda)$,

$$S_e(\lambda) \approx \left[\sum_k a_k T_k(\lambda) \right] \otimes \mathcal{G}(v; V, \sigma) + (\text{additive Legendre poly, deg 15--29}),$$

and $\sigma_e \equiv \sigma$, the fitted LOSVD dispersion. Because superposing spaxel spectra at different line-of-sight velocities V_i broadens the co-add, the width σ of that single fit *equals* the luminosity-weighted second moment $\sqrt{\langle (V_i - V_{\text{sys}})^2 + \sigma_i^2 \rangle}$ — so it measures the Cappellari 2013 eq. 29 / Gültekin 2009 eq. 1 quantity without ever forming the explicit sum (and with V_{sys} absorbed automatically into the fitted V ; §2.4.1). Code: `extract_aperture_spectrum` (the weighted sum) + `bootstrap_ppxf.setup_ppxf_inputs_from_spectrum` + `ppxf(..., moments=2)`. The explicit discrete sum $\sum_j F_j (V_j^2 + \sigma_j^2) / \sum_j F_j$ is computed only in the §7 annular cross-check (`run_gultekin_mc`), never in the headline.

- Why the integral path is the headline (not the discrete sum): the discrete spaxel-sum is (i) per-spaxel-S/N-limited and (ii) sensitive to arc contamination in the outer spaxels — an *unfiltered* discrete sum over the resolved PowerBins out to $\sim 3.4''$ is dominated by the $z=1.302$ arc (outer bins carry $\sigma \gtrsim 300\text{--}440$ km/s, V swings of $\pm 150\text{--}270$; the σ_e blows up to ~ 350). The I-weighted aperture auto-suppresses the arc (bright deflector core dominates the weight) and avoids the S/N floor, so it implements the same Cappellari definition robustly. The discrete §7 sum is therefore arc-filtered to $R < R_{\text{safe}}$ and used only as the architecture-independent cross-check / for the $\sigma(R)$ profile.
- Standard wavelength range (headline): rest 3800–5400 Å (`wR3800_5400_arcmask`) — Ca H+K through Mg b / Fe5270 / Fe5335; 2161 good pixels ($\sim 2.6\times$ the narrow window). The fit is constrained by the highest-S/N features — Ca H, Ca K, G-band (3800–4400) — and is consistent across windows.
 - Blue edge 3800 Å = XSL template floor (def-rest 3675 Å) + 5% velocity padding + retaining Ca H+K; red edge set by KCWI coverage. Provenance is data-driven, NOT a specific paper. Closest precedent: LEGA-C (Bezanson et al. 2018; van der Wel et al. 2021) at $z\sim 0.7$. DO NOT cite “Cappellari 2017 recommends extending blueward” — that was a hallucination, retracted.
 - Cross-check window: rest 4000–5400 Å (`wR4000_5400_arcmask`, H β + Mg b, no Ca H+K) — orthogonal Lick-tradition feature set.
 - Narrow legacy window: obs 6500–7500 Å = rest 3879–4476 Å (Ca H+K + G); $\sigma_e = 267.95 \pm 30.10$.

A few systematics tests to describe (full budget in Results §3.1): I-shape (10 I-weight-map shapes, ± 2.27), F200 mask weight ($w \in \{0, 0.5, 1\}$, ± 6.65), fit-window (3 windows, ± 3.82), centering (5-center $\pm 0.4''$ sweep, ± 4.0), reduction-pass (NEW vs OLD cube, ± 3.45). Plus the M9 “DO NOT MASK” finding: the $+5\text{--}7\%$ bump at def-rest 5193–5204 Å is the Mg b LOSVD wing (signal, not contamination); masking it drops σ_e by 7 km/s.

§2.3 Lens Modeling

G1 — now sourced from the companion-paper draft (2026-06-12). The lens-modeling content below is summarised from the Ferrami et al. draft (G. Ferrami, MPA Garching; lead-author email gferrami@mpa-garching.mpg.de), “A supermassive black hole in the Einstein spiral DES J0206–0114”, file 7Documents/AGEL/AGEL_0206_ApJL_Figures/SMBH_in_Einstein_Spiral_DESJ0206_0114_June12/main.md. This is a live draft with explicit PLACEHOLDER UNTIL FINAL RUNS ARE COMPLETED flags on the evidence table, the $M_\bullet$ – σ figure, and the potential-correction figure — every numeric BH/evidence value here is provisional. The complementary on-hand pointer: PROJECT_BRIEF.md notes the lens-model enclosed mass combines with $(\sigma, M_\bullet, z)$ to give a 4D constraint on the central BH + bulge scaling relations. *This is OUR (kinematics+photometry) paper; the lens model is the companion paper — cite it, do not reproduce its results as ours.*

⚠ Cross-paper reconciliation flags (resolve before any joint number is quoted):

- Cosmology — RESOLVED 2026-06-12: we adopted Planck 2015 to match Ferrami (Planck Collaboration 2016): $H_0 = 67.7$ km/s/Mpc, $\Omega_m = 0.302$. (Previously our paper used $H_0 = 70$, $\Omega_m = 0.3$.) Impact on our numbers is small and documented in the headline “Cosmology switch — impact” box: R_e 14.76→15.26 kpc, $\log M_\bullet$ 11.47→11.50 (+0.028 dex, exact D_L^2 rescale), σ_e and all angular quantities unchanged. Now both papers share a cosmology → a Ferrami $M_\bullet$ and our $M_\bullet/\sigma_e$ can be co-plotted directly. The Ferrami draft still carries its own standing TODO “Check all mass values to be corrected for the different deflector redshift” — that is the redshift (not cosmology) reconciliation below, and our z is the canonical one.
- Redshifts differ slightly. Ferrami quotes $z_l = 0.675$, $z_s = 1.303$; our systemic values (notebook 04) are $z_l = 0.67564$, $z_s = 1.30263$. Minor, but the $\sigma_{\text{inst}}/V_{\text{sys}}$ frame and the $M_\bullet$ conversion both use z_l — reconcile to our line-fit values when combining.
- System identity (agreed): AGEL J020613–011417 = DES J0206–0114, AGEL DR2 (Barone et al. 2026); ICRS RA = 31.55611°, Dec = –1.23817°. Discovered in DES via a CNN (Jacobs et al.). HST WFC3 F200LP + F140W, SNAP program 16773 (PI Glazebrook), two 300 s exposures per filter — the same imaging this paper uses for photometry. The lens forms a radial arc (three star-forming source galaxies near a fold + a cusp); the shallow, sub-isothermal total slope is what makes the radial image bright.

§2.3.1 Modelling code — **pronto**

- **pronto** (Vegetti & Koopmans 2009; Rybak et al. 2015; Rizzo et al. 2018; Ritondale et al. 2019; Powell et al. 2020; Ndiritu et al. 2025). Simultaneously recovers a resolved background source (regularised on a Delaunay adaptive tessellation) and the non-linear parameters of the smooth deflector light + mass distributions. The smooth potential can additionally be perturbed by regularised pixellated potential corrections (a.k.a. *gravitational imaging*). Hierarchical Bayesian framework; evidence $\ln Z$ computed by exploring the posterior with MultiNest (Feroz et al.; importance-nested-sampling variant). [Ferrami draft §Method]

§2.3.2 Mass density profiles

- Main lens = main elliptical parametric profile + mass multipoles + 2 secondary SIS deflectors + external shear. Three trial forms for the main elliptical component:
 - sEPL — singular elliptical power-law (the cEPL limit $r_c \rightarrow 0$). *This is the main halo of the best model.*
 - cEPL — cored elliptical power-law: $\kappa(\xi) = \kappa_0 \cdot (4 - \gamma) / (4\sqrt{q}) \cdot (r_c^2 + \xi^2)^{((1 - \gamma)/2)}$, $\xi = \sqrt{(x^2 + y^2)/q^2}$ the elliptical radius, r_c the core radius, γ the 3D slope, q the axis ratio. Deflection via **fastell** (Barkana). Einstein-radius definition for the cEPL from Enzi et al. (cEPL θ_E).
 - bEPL — broken elliptical power-law: piecewise κ with break radius θ_B , break convergence κ_B , inner/outer 2D slopes t_1/t_2 (O’Riordan, BPL derivation).
- Two companion galaxies within $\sim 4''$ of the main lens → singular isothermal spheres (SIS) (cEPL with $q=1$, $\gamma=2$, $r_c=0$).
- Multipoles of order $m = 1, 3, 4$, *circular slope-matched*: $\kappa(\theta, \varphi) = \theta^{(1 - \gamma)} \cdot [a_m \sin(m\varphi) + b_m \cos(m\varphi)]$. Choice of circular (not elliptical-isothermal) multipoles justified via Pagnat et al. (2025) eq. B10: at the recovered $\gamma \approx 1.3$, $q \approx 0.7$, circular multipoles are the better approximation for θ outside the band $(1 \pm 0.15) \cdot \theta_E$ — and the science target is the BH perturbation at $\theta \ll \theta_E$.

- External shear of strength Γ and PA θ_Γ .

§2.3.3 Central perturber (the BH test)

- After fixing the smooth-model parameters, the draft tests whether the data require an extra concentrated central mass by comparing Bayesian evidence across perturber choices:
 - Point mass (PM) — $\kappa(\theta) = (\theta_{\{E,\bullet\}}^2/2) \cdot \theta/|\theta|^2$, $\theta_{\{E,\bullet\}}$ the Einstein radius of a point mass $M_\bullet$ at z_l . Run twice: (i) fixed to the centre of the foreground mass, and (ii) free within a $0.5''$ radius of the light centre.
 - SPL — spherical power-law (EPL with $q=1$, $r_c=0$).
 - NFW — Navarro–Frenk–White with free concentration c ($\kappa_s = \rho_s r_s \Sigma_c$). The PM is the SMBH hypothesis; SPL/NFW are the extended-mass (dark-subhalo) alternatives it is tested against. [Ferrami draft §Method → Central black hole]

§2.3.4 Lens light + source regularisation

- Lens light = a single Sérsic with $m = 1, 3, 4$ multipole perturbations, fit *simultaneously* with the lens + BH mass in *pronto*. Evidence comparison disfavoured both a 2nd/3rd Sérsic and the no-multipole model. During burn-in, 3 sets of multiple images are required to focus to a tolerance shrinking to $0.05''$.
- Source = Delaunay adaptive tessellation with adaptive, SNR-weighted regularisation (O’Riordan, Euclid lens): two forms tested — penalising the gradient vs the curvature of the source surface brightness; per-pixel weight $\log(\lambda_i/\lambda_s) = -w_a x_i^{(-w_b)}$, x_i the log-SNR normalised over all pixels, with w_a, w_b free. Curvature regularisation wins by $\Delta \ln Z \approx 100$ over gradient → the paper uses curvature throughout. [Ferrami draft §Adaptive source regularisation]

System identity for the summary sentence: AGEL J020613–011417 (DES J0206–01), AGEL DR2; source $z = 1.302$ (spiral), deflector $z = 0.67564$.

§2.4 End-to-end procedures (ordered walkthroughs)

*This section spells out the actual step order of each pipeline — what runs, in what sequence, with which script — so the Methods section and a referee can reconstruct every headline number. Numeric results are the M12 headline (*results/PAPER_VALUES.json*).*

§2.4.1 Spectroscopic σ_e pipeline — step by step

Driver for the headline: `scripts/run_wide_sigma_e.py --cube new_clean_hei --n_bootstrap 500` (calls `final_sigma_e.load_setup` → `extract_aperture_spectrum` → `bootstrap_ppxf` machinery).

1. Reduction (upstream). Keck II / KCWI, KCRM red arm, Red-Low grating, Medium slicer, 2×2 binning, dithered PA $0/45/90^\circ$. The headline cube is the `NEW_mtwdc_reduction` (Master-Twilight+Dome hybrid flats, K. R. Gupta), `Nov17_2025_DESJ0206_RL_combined_mtwdc_icubes_wcs.fits`. The `pre_mtwdc` cube is the OLD reduction (second-reduction cross-check). Cube axes $[\lambda, y, x]$; $0.30''/\text{spaxel}$; seeing FWHM $1.27''$ (GUIDFWHM); $\sigma_{\text{inst}} \approx 12.6 \text{ km/s}$; $R \approx 10000$.
2. Geometry + R_e . HST-mean centre = `photutils.centroid_2dg` of the F140W+F200LP cores (offset $0.36''$; F200LP pulled by the UV arc, so F140W is the clean bulge centroid), propagated through the KCWI WCS to a sub-pixel IFU centre. R_e = mean of the F140W ($1.912''$) + F200LP ($2.281''$) best-mask curves-of-growth = $2.097''$ (`final_sigma_e.curve_of_growth`, `r_max=6''`; `photutils-validated` $\pm 0.002''$).
3. Spatial arc mask → IFU grid. The `F200LP_mask.fits` (2512 HST px, arc-only) is reprojected onto the $0.30''$ IFU grid with `scipy.ndimage.map_coordinates(order=0)` (nearest-neighbour) → `arc_spax_mask`. Mask coverage within R_e (PSF caveat; `scripts/psf_mask_fraction.py`): this *geometric* mask flags $35/154 = 23\%$ of spaxels (28% by I-weight) inside $R_e=2.097''$. But the arc light is spread by the $1.27''$ KCWI seeing, so the geometric fraction is a lower bound: growing the mask by the PSF (binary dilation) raises the masked fraction to 71% / 81% (by $\text{FWHM}/2 = 0.64''$) or 96% / 98% (by the full $\text{FWHM} = 1.27''$)

- i.e. a hard seeing-grown mask removes nearly the entire aperture and is infeasible. (Convolving the binary mask with the PSF and cutting at $>25\%$ / $>10\%$ arc-light contamination gives 42% / 75% by count; the $>50\%$ cut, 1% , is a thin-arc dilution artifact.) This is precisely why we instead (i) I-weight the aperture — the bright deflector core dominates, the arc-contaminated outskirts are down-weighted — and (ii) carry the mask-weight systematic (± 6.65 km/s, $w=0 \rightarrow 1$; §2.4.3) to bracket the residual arc dilution rather than attempt a hard PSF-grown mask.
4. I-weighted aperture extraction. Build the 6500–7500 Å white-light I-map; the $R < R_e$ aperture spectrum = $\Sigma_{\text{spaxel}} \text{cube} \cdot w$, w = I-weight with arc spaxels hard-masked ($\text{mask_weight}=0$) and re-normalised. Spectra are summed (not averaged) so the bootstrap preserves per-spaxel noise. (`extract_aperture_spectrum`, `final_sigma_e.py`.)
 5. Spectral masking (goodpixels). (a) no-Balmer mask — forbidden lines only ([O II], [O III], [O I], [N II], [S II]); H δ , H γ , H β kept (absorption; M12/nb16 confirmed keep-unmasked).
 (b) $z=1.302$ source-emission masks ARC_MASKS_REST (Mg II, [O II], [Ne III], He I 3819) + the O $_2$ A-band telluric. (c) BAD_PIXELS_REST = 35 entries (26 CR residuals + 9 M10 OH/sky bands). All in deflector rest-frame Å.
 - Outlier rejection / sigma-clipping decision: ppxf’s in-fit `clean=False` — `clean=True` rejects 0 pixels here (the KCWI noise array is overestimated vs the actual residual scatter), so it is inert. Instead, cosmic-ray/bad pixels are flagged by a 3σ local-MAD clip (rolling 75-pixel window, $|\text{residual}|/\text{local-MAD} > 3\sigma$) on the canonical residuals $\rightarrow$ 52 CR-like spikes, curated into the 26 CR entries of BAD_PIXELS_REST; the 9 M10 sky bands were flagged separately at $>2.5\times$ median sky-noise. Caveats (carried as a stated sensitivity, not a budget term): the 3σ threshold is not strongly motivated vs alternatives, and cleaning shifts σ_e by only $+1.4$ km/s (below stat 1σ); the same pixels replicate the shift on the OLD cube (M6) $\rightarrow$ real CR residuals.
 6. Per-SPS frame-aware ppxf. For each of FSPS / EMILES / XSL: match the galaxy frame to the SPS native frame (FSPS=vacuum, EMILES/XSL=air; scalar-median $\text{vac} \leftrightarrow \text{air}$ via Ciddor 1996), run ppxf (Cappellari) with `moments=2`, additive Legendre `degree` swept 15–29 (15 values), `mdegree=0`, over the wide window (rest 3800–5400 Å). Subtract the per-SPS V_{sys} before pooling.
 7. Wild-bootstrap errors. Per (SPS $\times$ degree), $N=500$ hybrid wild bootstrap (`bootstrap_ppxf.compute_local_residual_scaling`): residuals $r = \text{galaxy} - \text{bestfit}$; a rolling 75-pixel window estimates the local (wavelength-dependent) noise scale $s(\lambda)$ (clipped to $[0.2, 5.0]$); each iteration applies a Rademacher ± 1 sign-flip to the locally-rescaled residual, adds it back to the best fit, and re-fits $\rightarrow$ one σ draw. Seeds are deterministic (`BOOT_SEED=42 + 50000 + 10000 \cdot W\_IDX + 100 \cdot \text{sps\_idx} + \text{degree\_i}`) so caches bit-reproduce. `joblib` `loky`, BLAS pinned to 1 thread/worker; $N=50$ smoke before $N=500$ production.
 8. Pool + budget. Concatenate all σ samples (3 SPS $\times$ 15 degrees $\times$ 500) $\rightarrow$ 16/50/84 percentiles = central + asymmetric stat. The pooled width already marginalises SPS + polynomial degree. Add the §2.4.3 systematic components in quadrature $\rightarrow \sigma_e = 267.31 - 11.98/+11.58$ (sym ± 11.77) km/s (at the best-mask $R_e=2.097''$ aperture; cache `results/run_wide_sigma_e/resys_best_mean/`).

§2.4.2 Photometry / M★ pipeline — step by step

Drivers: `scripts/photometry_systematics.py` (masking + photometry), `bagpipes_sersic_refit.py` (SED), orchestrated/displayed in notebook 12. Recipe = F200LP locates, IR extends, fill-in recovers.

1. Locate the arc on F200LP (best source contrast): a 2D Sérsic deflector model is subtracted and a pixel is flagged as arc where its positive residual exceeds $k \cdot \sigma_{\text{sky}}$ — i.e. k is the detection threshold in units of the sky noise σ_{sky} (so $k=3$ means residual $> 3\sigma_{\text{sky}}$), with a parallel S/N floor (the pixel’s own $S/N > \text{SNR_thresh}$). We adopt $k = 3$ (`arc_mask_verification.py`): this objective Sérsic-residual mask reproduces the expert hand mask to 0.016 mag, R_e to 3%. $k = 3$ is the clean saturation point of a 2-D sweep over the two thresholds ($\text{SNR_thresh} \in \{2..20\} \times k \in \{2..8\}$) — below $k \approx 3$ it grabs noise, above it the masked-flux fraction stops changing.
2. Reproject the F200LP footprint to every band (WCS + `map_coordinates`) — reproduces the expert JWST aperture mags to 0.01–0.02 mag.
3. IR-extend with a COLOR GATE. In the deeper/sharper JWST bands, region-grow the arc seed into pixels that are (a) significant positive residual above the single-Sérsic deflector model ($\text{data} - \text{model} > 3\sigma_{\text{sky}}$), (b) bluer than the deflector core by ≥ 0.5 mag in F200LP–F140W, and (c) contiguous with the seed within

- $R < 5''$. The color gate (b) is essential: a single Sérsic slightly under-fits the bright IR body, so residual-only growth would sweep the *red* galaxy body and companions into the mask; the *blue* $z=1.302$ source vs *red* passive deflector contrast keeps only genuine source pixels. See §2.1.1b for the detailed rationale + caveats.
4. Deflector model = single-component Sérsic — physically motivated for the elliptical (per user 2026-06-10); not a bulge+disk decomposition. Fit gotcha: `Sersic2D.amplitude` is $I(r_{\text{eff}})$ not the peak — `init ~ peak * exp(-b_n)`, bound to $(\text{peak} \cdot 1e-3, \text{peak} \cdot 1.5)$, else `LevMar` collapses to ≈ 0 . PSF-convolved fill quantified at ≤ 0.004 mag (`psf_fill_model.py`) — negligible.
 5. Two photometry estimators per band/mask: raw `photutils.aperture` (masked $\rightarrow$ discards under-arc deflector light $\rightarrow$ biased low, mask-size-dependent) vs single-Sérsic fill-in (replace masked pixels with the model $\rightarrow$ recovers it). Fill-in correction -0.19 to -0.27 mag under the color-gated masks (smaller than the old 2-component $+0.18$ – 0.96 mag — cleaner masks remove less).
 6. AB magnitudes from per-image header zeropoints (never hardcoded): HST PHOTFLAM/PHOTPLAM, JWST PIXAR_SR. Four bands: F200LP, F140W (HST WFC3), F150W2, F322W2 (JWST NIRC2).
 7. Bagpipes SED (`02_streamlined_Bagpipes_SED`): exponential- τ SFH, Calzetti dust (A_V 0–2), free Z (0–2.5 $Z_\odot$), z prior (0.674, 0.676); Nautilus sampler $n_{\text{live}}=400$, 500-sample posterior. A 10% fractional flux floor per band covers photon noise + the drizzle pixel-correlation under-estimate (empirical noise ≈ 1.3 – $2 \times$ CCD-equation); 20% is run as a conservative variant.
 8. $M_\star$ budget = 8 fits {per-band, global} $\times$ {raw, filled} $\times$ {10%, 20%} (script-reproducible via `scripts/mstar_masking_budget.py` $\rightarrow$ `Mstar_masking_systematic.npz`, `photometry_systematics_Mstar.npz`) $\rightarrow$ headline (empirical, raw-central with the best per-band color-gated mask, one-sided +sys, user choice): $\log M_\star = 11.36 \pm 0.08$ (stat) $+0.14$ (sys) at 10% [11.26 ± 0.14 $+0.18$ at 20%], fill-in reach 11.47; explicit masking-approach systematic ± 0.086 dex (was ± 0.16 under the 2-component masks). Central consistent with the validated expert hand-mask 11.33.

§2.4.3 Systematic-audit procedures (how each budget term was measured)

Each σ_e systematic is a dedicated sweep re-using the §2.4.1 machinery, varying ONE axis and quoting peak-to-peak/2 (or half- Δ for two-point axes), all on the headline cube + masks:

- I-shape (± 2.27): 10 alternative I-weight maps (`run_isource_shape_sweep.py`) $\times$ 3 SPS $\times$ 15 deg $\times$ $N=250$; peak-to-peak/2 of the 10 central σ_e (266.83–271.37).
- F200 mask-weight (± 6.65): down-weight the arc spaxels at $w \in \{0, 0.5, 1\} \times 3 \text{ SPS} \times N=500$; ($w_{00} 269.69 - w_{100} 256.39$)/2. (*This is the mask-strength axis. The mask-definition axis — expert/sersic/perband/global reprojected to the IFU grid, `run_sigma_e_mask_systematic.py` — gives ± 4.58 , which overlaps this; larger-of-two is kept, not added.*)
- Fit-window (± 3.82): 3 windows ($w_{R3800_5400} / w_{R4000_5400} / w_{6500_7500}$) $\times$ 3 SPS $\times$ $N=500$; peak-to-peak/2 (269.66 / 268.58 / 276.22).
- Frame (vac/air): DROPPED 2026-06-14 — deterministic per-SPS native-frame correction (< 0.5 km/s σ impact, TESTS D4); the former ± 5 was the inter-SPS spread already in the pooled stat (double-count).
- Centering (± 4.0): 5 perturbed HST-mean centres ($\pm 0.4''$ sweep; `NOTES_centering_investigation`).
- Reduction-pass (± 3.45): half- Δ between the NEW and OLD cleaned cubes (269.62 vs 262.72); 2 reductions only — refine if a 3rd lands.
- R_e -source (± 5.13): σ_e re-measured over a 7-point R_e grid bracketing the $2.097''$ best-mask headline (`run_sigma_e_Re_grid.py`). Adopted = best-mask CoG family $\{1.912/2.097/2.281''\}$ peak-to-peak/2 = $(269.99 - 259.73)/2$. The full grid incl. CaHK+G $2.90''$ (± 9.98) and CaHK+G's $+12.4$ deviation are cross-checks, NOT folded (CaHK+G is a different I-map definition, not an error on the photometric R_e). Was ± 6.13 at the old expert-mask headline.

Quote the budget table (§3.1) as the result; the iterative history that *led* to this masking + component set is the internal audit trail (M8 $\rightarrow$ M12) at the end of this document — not for the manuscript Methods section.

§2.4.4 Reproducibility / infrastructure

- Architecturally-independent estimators (must not share intermediate state): §6cum cumulative I-weighted single-ppxf aperture (the Cappellari 2006 co-add-then-fit method = headline) $\cdot$ §7 discrete Gültekin (2009)

eq. 1 annular spaxel-sum (`run_gultekin_mc`, arc-filtered to $R < R_{\text{safe}}$) · nb07e arc-spectrum subtraction. The three agree at $<1\sigma$ (267.32 co-add-fit vs 256.17 discrete, narrow).

- Caching + seed discipline. Every expensive bootstrap writes a `.npz` keyed by its parameters (`results/<sweep>/<params>_N{N}.npz`); a cached result is never re-run. Seeds are deterministic (§2.4.1 step 7) so re-runs bit-reproduce; $N=50$ smoke before $N=500$ production; joblib with `BLAS=1/worker` to avoid over-subscription.
- Deterministic values registry (single source of truth). `scripts/paper_values.py` recomputes every headline number from the result caches (with per-entry provenance + formula) → emits `results/PAPER_VALUES.json`. The ApJL figures load that JSON (no hard-coded literals); the summary docs are validated by `paper_values.py --check <files>`, an anchored, precision-aware drift linter (parses only the canonical $\sigma_e = N \pm M / \log M \star = N$ statements; dated snapshots carry a `pv-skip-file` marker). Regenerate + verify: `python scripts/paper_values.py && python scripts/paper_values.py --check *.md`.

§3 Results

Figure 2 (left panel) — the integrated spectrum + SED

- Left: KCWI/KCRM Red-Low integrated, $I(r)$ -convolved spectrum of the deflector elliptical, with the ppxf best fit overlaid (and inline residuals — Rodrigo’s standard nested-gridspec style). Loads from the wide-window fit + bootstrap pool.
- Right: HST+JWST four-band SED + Bagpipes posterior model. Loads `results/bagpipes_sed_results.npz`.
- Figure file: `AGEL0206_sigma_e_SED_final_wide.pdf` (per outline) / `results/AGEL0206_spectra_SED_fit.pdf`.

§3.1 Velocity Dispersion Measurement

- Final redshifts (to quote in Results). The deflector systemic redshift from Gaussian line fits (6 absorption features, §2.2.2) is $z = 0.67564 \pm 0.00033$, in agreement with the independent ppxf stellar-kinematic redshift $z = 0.67555 \pm 0.00002$ ($V_{\text{sys}} = -16 \pm 4$ km/s relative to the line-fit value; headline $R_e = 2.097''$ aperture). All three SPS libraries converge: FSPS 0.67555 / EMILES 0.67553 / XSL 0.67556 (~4 km/s spread) — the signature of the air↔vacuum per-SPS frame fix (§2.4.1), which collapsed the legacy ~110 km/s FSPS-vs-air offset. The 16 km/s line-fit↔ppxf offset is well within the ± 59 km/s line-fit uncertainty (and the ppxf formal ± 4 km/s carries an additional $\sim \pm 5$ km/s per-SPS frame systematic). We adopt $z = 0.67564$ (the line-fit value supersedes AGEL DR2 $z = 0.67511$). The lensed-source redshift is $z = 1.30263 \pm 0.00003$ (red-cube [O II] $\lambda\lambda 3726, 3729$ doublet). [§2.2.2; `scripts/redshift_verify.py`, `run_sigma_e_Re_grid.py`]
- Headline: $\sigma_e(<R_e) = 267.31 \pm 11.77$ km/s (asym $-11.98/+11.58$), the luminosity-weighted σ_e within R_e (Gültekin et al. 2009 eq. 1 definition) measured by the Cappellari et al. 2006 (SAURON IV §2.3) co-add-then-fit method — a single ppxf fit to the $I(r)$ -weighted $R < R_e$ aperture spectrum ($R_e = 2.097''$ best-mask CoG) — plus the wild-bootstrap error pool. Supersedes 269.62 ± 13.27 at $R_e = 2.305''$ — the “best mask throughout” cascade (2026-06-11) adopts the best-mask $R_e = 2.097''$, lowering σ_e by 2.3 km/s along the rising $\sigma(R)$ profile (well within errors).
- Reproduce: σ_e at the headline aperture is `results/run_wide_sigma_e/resys_best_mean/(R_e=2.097'')`, produced by `scripts/run_sigma_e_Re_grid.py --n_bootstrap 500`; the R_e -source systematic (± 5.13 , best-mask CoG family) is from the same grid (`results/sigma_e_Re_grid_N500.npz`).

R_e measurement and checks (for the §3.1 paragraph): $R_e = 2.097'' = 14.76$ kpc, the F140W (1.912'') + F200LP (2.281'') best-mask-CoG mean (photutils-validated $\pm 0.002''$); supersedes expert-mask 2.305'' and IFU-only 2.61''; masked CoG removes arc/companion bias.

σ_e systematic budget — *generated by `scripts/paper_values.py --render`; do not hand-edit inside the markers.*

Component	$\pm$ km/s	Note
stat (N=500)	4.51	asym $-5.04/+3.98$; pooled 3 SPS $\times$ 15 deg (marginalizes SPS + degree)
I-shape	2.29	14-shape peak-to-peak/2 (266.79-271.37): the 10 raw maps + 4 arc-free PSF-matched (Sérsic/filled $\rightarrow$ 1.27" conv); was ± 2.27 over 10
arc masking	6.65	arc masking: (w00-w100)/2 weight sweep; subsumes mask-approach 4.58 (larger-of-two)
centering	4.00	carried constant
fit-window	3.82	peak-to-peak/2 across 3 fit windows
reduction-pass	3.45	half- Δ between NEW and OLD reductions
R_e-source (best-mask grid)	5.13	peak-to-peak/2 across best-mask CoG family {F140W 1.91", mean 2.10", F200LP 2.28"} at the 2.097" headline (user 2026-06-11; CaHK+G & full grid are cross-checks, not folded)
TOTAL (sym)	11.77	quadrature (sys $\pm 10.87 \oplus$ stat)
TOTAL (asym)	$-11.98 / +11.58$	preserves stat-side skew

Cross-checks (not added): arc-mask-definition ± 4.58 (overlaps F200 mask) $\cdot$ full-grid R_e-source ± 9.98 (incl. CaHK+G, conservative ceiling) $\cdot$ CaHK+G(2.90") deviation $+12.38$ km/s.

(Per-component sweep detail — the per-window / per-estimator values behind each $\pm$ — is in §2.4.3.)

- No separate SPS line is folded in: the SPS-library spread collapsed from ~ 26 km/s at the narrow window (FSPPS 253.0 < EMILES 267.5 < XSL 279.7) to ~ 4 km/s at the wide window (FSPPS 271.3 / EMILES 267.2 / XSL 270.4; PAPER_VALUES.json:per_sps_p50) — a key argument for the wide window; it now sits inside the pooled stat width (between-SPS ± 2.04).
- Quadrature is deliberately conservative (components not strictly independent).
- R_e-source systematic IS now folded in (± 6.13 , M12 2026-06-08): re-measured at the wide window (nb15) and included in the budget above — the narrow-window ± 16.9 is superseded. The full 4-estimator spread was adopted (user decision); the CaHK+G 2.90" estimator drives it ($+10$ km/s = rising $\sigma(R)$, see below), while the light-R_e family alone gives ± 2.50 .

Systematics paragraph order (per outline): first R_e and I(r) (I-shape ± 2.27 , R_e-source flagged), then the spectrum (mask ± 6.65 , window ± 3.82 , reduction ± 3.45 , centering ± 4.0). (The vac/air frame term, formerly ± 5.0 , is not carried — it is a deterministic per-SPS correction with <0.5 km/s σ impact, and its inter-SPS residual is already in the pooled stat; dropped 2026-06-14.)

Cross-checks (narrow window, architecturally independent, superseded for headline but valid):

- §6cum cumulative I-weighted aperture ppxf: 267.32 ± 24 ; $\sigma_e(<R_e/2) \approx 225.78$; $\sigma_e(<R_e/8) \approx 209.18$.
- §7 discrete Gültekin annular sum (equal-N inside $R_{\text{safe}}=3R_e/4$): $256.17 -13.0/+12.7$.
- §7b flat- σ extrapolation: $274.37 -16.2/+17.4$.
- nb07e arc-spectrum subtraction matches §6cum bit-identically (<1 km/s).
- Narrow Ca H+K + G window (nb09d): 267.95 ± 30.10 — headline within ~ 2 km/s.

Headline evolution (mask-audit arc): 268.98 (clean) $\rightarrow$ 271.87 (+He I, M8) $\rightarrow$ 269.62 (+M10 sky audit) $\rightarrow$ ± 11.77 (M11 re-derivation) $\rightarrow$ ± 13.27 (M12, +R_e-source ± 6.13 , 2026-06-08). He I pushes σ up ~ 2.9 , the 9 sky bands pull it down ~ 2.2 (near-cancel); central value 269.62 stable across M10 $\rightarrow$ M12 (only the error grew).

Compare with Greene (2016/2020) and KH13 choices: $\sigma_e(<R_e)$ here uses the luminosity-weighted single-aperture σ_e (Cappellari et al. 2006 SAURON IV §2.3 co-add-then-fit; Gültekin 2009 eq. 1 definition), exactly as adopted by Kormendy & Ho 2013 (M $\bullet$ - σ eq. 3) and Greene et al. 2020 ARA&A (fig. 5). We dropped $R<R_e/8$ (JFK95 σ_c) because seeing under-resolves it. For a pure elliptical, galaxy $R_e \approx$ bulge R_e , so KH13's "bulge R_e " equals the measured total R_e — no bulge/disk decomposition needed.

σ_e physical-meaning discussion: the deflector is a dispersion-supported elliptical bulge; $\sigma(r)$ should *decrease* gently with radius — any rising $\sigma(r)$ in raw output is a continuum-dilution artifact from the $z=1.302$ arc, not a physical gradient. Resolved kinematics (nb11): 17 central spaxels at $S/N \geq 5$, $\sigma \in [144, 251]$ km/s (median 201), declining with R , no coherent rotation above noise.

§3.2 Stellar Mass Estimate

HEADLINE: $\log(M_\star/M_\odot) = 11.5 \pm 0.1$ (stat) ± 0.2 (sys) (registry 2dp: $11.46 +0.07/-0.06$ stat ± 0.17 sys) (validated-fit aperture-corrected total, matched $2 R_e$, 10% flux floor; Planck 2015).

Final $M_\star$ procedure (2026-06-15 audit)

The total-light stellar mass is built in two stages — photometry, then SED fit — each pinned by an audit this session that superseded the earlier 11.50 (the old value was right only by luck: two offsetting errors).

1. Total-light photometry — validated-fit aperture correction (`scripts/aperture_correction_validated.py`). Measure the empirical flux in a matched $2 R_e$ elliptical aperture (same physical region in all 4 bands; color/morph-gated, WCS-registered arc mask + companion mask), then correct to total by (a) filling the masked pixels and (b) adding the beyond-aperture wings, using a single-Sérsic model whose shape (n , R_e , ellipticity, PA) is fixed to the photutils-validated per-band fit and whose amplitude+sky are fit to the data. This replaces the earlier `auto_fit_sersic`, which railed to $r_{\text{eff}} 2.2\text{--}2.7''$ (vs the validated $1.6\text{--}2.0''$ and the CoG $R_e = 2.097''$) with ellipticity $\rightarrow 0$, over-correcting the beyond-aperture light by ~ 0.15 mag/band and inflating the old total to 11.50. The fixed correction drops the total-light photometry by ≈ 0.11 dex. Method refs: Taylor+2011 (GAMA aperture $\rightarrow$ total), Sonnenfeld+2013 (lens-deflector Sérsic photometry), Graham&Driver2005 (Sérsic total formalism).
2. Spectrum-consistent SED fit — quiescence-constrained prior (`scripts/mstar_headline_quiescent.py`). The 4-band SED is degenerate (age–dust– M/L): an old+passive and a young+dusty solution fit the colours equally well ($\Delta \ln Z = -0.18$, identical χ^2 ; `scripts/sed_quiescence_check.py`), and the unconstrained flat-age Bagpipes prior slides onto a spurious young+dusty branch (SFR $\sim 57 M_\odot/\text{yr}$). The KCWI spectrum — strong Ca H&K / G-band / Mg b / Fe absorption with no emission lines — decisively selects the old, passive solution. The headline therefore adopts a quiescence-constrained SFH prior (old age, short τ , low dust); the fiducial $\leftrightarrow$ quiescent spread is carried as the SFH-prior systematic. The old population’s higher M/L (+0.10 dex) nearly cancels the photometry drop (-0.11 dex), so the headline returns to ≈ 11.46 — close to the old 11.50, but now correct on both the photometry and the stellar population.

How the quiescence-constrained prior is defined (`scripts/mstar_headline_quiescent.py`). Both fits use the same model family — a single exponentially-declining (“delayed-free” τ) SFH, $\text{SFR}(t) \propto \exp(-t/\tau)$; Calzetti dust; free metallicity; redshift bracketing the line-fit z — and the same prior shape (every parameter has a uniform / top-hat prior in Bagpipes, set by a (l_o, h_i) tuple). The quiescent prior is not a different model or an informative (Gaussian) prior; it is the fiducial prior with the bounds of three parameters tightened to the region the KCWI absorption-line spectrum independently confirms, so the degenerate young+dusty branch is excluded from the prior volume:

Bagpipes parameter	fiducial (flat)	quiescent (headline)	why this enforces quiescence
<code>exponential:age</code> (Gyr)	(0.1, 15)	(4, 15)	age of the <i>onset</i> of star formation. Lower bound 4 Gyr (vs the ≈ 7.4 Gyr age of the Universe at $z=0.676$) forces an old population formed at $z \gtrsim 1.5\text{--}2$; kills the few-hundred-Myr young solutions
<code>exponential:tau</code> (Gyr)	(0.3, 10)	(0.1, 1.5)	SFH e-folding time. Short $\tau \leq 1.5$ Gyr $\rightarrow$ a fast-declining, early-quenched history; with age ≥ 4 Gyr the present SFR is suppressed by $\gtrsim e^{-(4/1.5)} \approx 0.07$ of peak $\rightarrow$ low sSFR. Long- τ (still-forming) histories are excluded

Bagpipes parameter	fiducial (flat)	quiescent (headline)	why this enforces quiescence
dust:Av (mag)	(0, 2.0)	(0, 0.6)	breaks the age–dust degeneracy: the spurious young solution needs $A_V \approx 0.75$ to redden a young SED into the observed red colours. Capping $A_V \leq 0.6$ removes that escape route, leaving age (not dust) to produce the red continuum
exponential:massformed (log $M_\odot$)	(1, 15)	(1, 15)	unchanged — total stellar mass formed
exponential:metallicity ($Z_\odot$)	(0, 2.5)	(0, 2.5)	unchanged
dust:type	Calzetti	Calzetti	unchanged
redshift	(0.674, 0.676)	(0.674, 0.676)	unchanged — brackets the line-fit $z=0.67564$

The three tightened bounds (old age $\oplus$ short τ $\oplus$ low dust) are the operational definition of “quiescent.” They are motivated by an independent dataset (the spectrum), not by the broadband photometry being fit, so they are not circular: the photometry cannot break the age–dust–M/L degeneracy on its own ($\Delta \ln Z = -0.18$, identical χ^2 between branches; `scripts/sed_quiescence_check.py`), and the spectrum says which branch is real. The fiducial $\leftrightarrow$ quiescent $M_\star$ spread (0.08 dex) is propagated as the SFH-prior systematic so the choice’s impact is carried in the error budget rather than hidden. Cross-check (drop-F200LP, 2026-06-17, `scripts/mstar_drop_f200lp.py`): refitting without the bluest band under the same quiescent prior gives $\log M_\star = 11.56$ (Planck), within the ± 0.17 sys budget of the 11.46 headline — $M_\star$ is robust to removing the near-UV lever arm. Under the *fiducial* prior the 3-band fit collapses even harder onto the young+dusty branch ($\text{SFR} \approx 56 M_\odot/\text{yr}$), reconfirming that the spectrum-motivated prior, not any single photometric point, is what selects the physical solution.

Five estimators (empirical $\rightarrow$ model), validated photometry + quiescent SED prior, at 2 R_e :

estimator	log $M_\star$	what it adds
raw (empirical aperture, masked)	11.28	nothing — lower bound
raw + apcorr (most-empirical total)	11.34	validated-model wings only
filled (within aperture)	11.42	validated-model fill of masked pixels
total (aperture-corrected) — headline	11.46	fill + wings
Sérsic-total (pure model)	11.49	everything modeled

The ladder is now monotonic and total agrees with the pure-Sérsic total to 0.02 dex. (The old auto pipeline had total = 11.50 sitting *above* the Sérsic-total 11.43 — the inversion that flagged the over-correction.)

Passive stellar population (quiescence-constrained fit): mass-weighted age 3.8 Gyr, SFR 8 $M_\odot/\text{yr}$, sSFR -10.5 (quiescent), Z 1.25 $Z_\odot$, A_V 0.21 mag. The unconstrained flat-age fit’s SFR ~ 57 / sSFR -9.6 is a degeneracy artifact ruled out by the spectrum — see §3.3.6.

$M_\star$ error budget — *generated by `scripts/paper_values.py --render`; do not hand-edit inside the markers.* The headline error is a single named-component quadrature (like σ_e): the Bagpipes statistical posterior combined with the masking, aperture-correction-model, and SFH-prior systematics. The Sérsic-only quadrature is retained below as an independent model-path cross-check.

(1) Headline — $\log(M_\star/M_\odot) = 11.46 = 11.5$ (validated photometry + quiescent, spectrum-consistent SED prior; matched 2 R_e , 10% floor):

Component	$\pm$ dex	Note
stat (Bagpipes posterior)	+0.069 / -0.057	quiescent prior — tight; the age–dust–M/L young-solution tail is removed by the spectrum-motivated prior
masking-approach	0.086	under-arc (raw $\leftrightarrow$ filled) 0.057 $\oplus$ mask-def (per-band $\leftrightarrow$ global) 0.028

Component	$\pm$ dex	Note
aperture-correction model	0.118	auto-fit vs validated-fit Sérsic shape (the over-correction this audit removed)
SFH prior	0.083	fiducial flat-age vs quiescent SED prior (the age-dust-M/L degeneracy the 4-band SED cannot break)
sys (quadrature)	0.168	masking $\oplus$ apcorr-model $\oplus$ SFH-prior
REPORTED	+0.07 / -0.06 (stat) $\pm$ 0.17 (sys)	one-decimal: 11.5 ± 0.1 (stat) $\pm$ 0.2 (sys)
CoG cross-check	11.69	empirical curve-of-growth total-light (cD envelope); lands +0.04 dex beyond $+1\sigma \rightarrow$ total-light ambiguity already captured by the apcorr-model + sys budget, NOT added separately

(2) Sérsic-only estimator ($\log(M_\star/M_\odot) = 11.49$) — independent model-path systematic envelope (cross-check):

Component	$\pm$ dex	Note
stat	0.121	Bagpipes posterior width
mask	0.125	arc-mask choice (expert $\leftrightarrow$ global Sérsic-total) — dominant
model-form	0.057	single-Sérsic vs alternative model form
flux-floor	0.034	10% $\leftrightarrow$ 20% flux floor
fit-param	0.029	Sérsic-n / fit-parameter spread
apcorr-recon	0.026	apcorr $\leftrightarrow$ pure-model reconstruction
TOTAL	0.19	stat $\pm$ 0.12 $\oplus$ sys $\pm$ 0.15 (mask-dominated)

Five $M_\star$ estimators (empirical $\rightarrow$ model, validated photometry + quiescent SED prior, matched 2 R_e , Planck 2015): raw 11.27 (empirical lower bound) $\cdot$ raw+apcorr 11.33 $\cdot$ filled 11.42 $\cdot$ total 11.46 (headline) $\cdot$ Sérsic-total 11.49.

(Per-component provenance — which cache + sweep produced each $\pm$ — is in the bullets below and in §2.1.1b / §2.4.3. The σ_e budget is in §3.1.)

- Sérsic-only (full-model) systematic budget = ± 0.19 dex (scripts/sersic_total_systematic.py; elliptical multi-start fit, 2026-06-11): stat $\pm 0.121 \oplus$ sys ± 0.147 , with arc-mask choice ± 0.125 dominant, model-form ± 0.057 , flux-floor ± 0.034 , Sérsic-fit-n ± 0.029 , apcorr $\leftrightarrow$ pure-model ± 0.026 . (Was ± 0.17 with the circular fit; the mask term grew because the elliptical model widens the expert $\leftrightarrow$ global Sérsic-total gap.)
- Apcorr chain validated vs established codes (2026-06-11, scripts/validate_apcorr_established.py; petrofit/statmorph absent $\rightarrow$ photutils + scipy incomplete-gamma references): (1) b_n Ciotti&Bertin99 vs exact $\text{gammaincinv}(2n, \frac{1}{2}) \leq 0.05\%$; (2) the Sérsic total-flux formula (Graham&Driver05 eq.4) vs a numeric render-to-20 $R_e \leq 0.03\%$; (3) the aperture correction — rendered-model enclosed light vs the analytic Sérsic incomplete-gamma law $\gamma(2n, b_n \cdot 2\{1/n\})/\Gamma(2n)$ (Graham&Driver05) — $\Delta \leq 0.0007$;
 (4) the empirical hard-edge aperture $\text{sum}(\text{in_aperture})$ vs photutils `EllipticalAperture(method='exact')` $\leq 0.09\%$; (5) the finite-cutout truncation of $\text{sum}(\text{model_full})$ (the apcorr denominator) vs the to- ∞ total $\leq 0.19\%$ (≤ 0.002 mag). No production bug; $M_\star$ unchanged.
- Single-Sérsic ellipticity fix (2026-06-11). The deflector light model is a single Sérsic per band fit on the best mask. The fitter originally had a single low-ellipticity start that railed into a spurious circular χ^2 minimum ($b/a=1$ for all bands), under-reporting the shape and biasing the (1-ellip) total flux. Fixed with a multi-start (flux-weighted-moment seed à la SExtractor/GALFIT
 - a PA grid + circular fallback, lowest-residual). It recovers the true shape; $M_\star$ headline (total) is unchanged at 11.47 (empirical-aperture-dominated). Validated by injection-recovery against astropy's peer-reviewed Sersic2D (scripts/validate_sersic_fitter_synthetic.py): clean recovery for the deflector regime ($n \approx 1.2$ – 1.6 , $b/a \leq 0.85$), with a realistic b/a uncertainty $\sim \pm 0.06$ (the formal bootstrap is too tight). GALFIT/imfit/petrofit/statmorph are not installed (conda-env policy).
- The flat-age $M_\star$ posterior had a long LOW tail — now broken by the spectrum (2026-06-15). Under the unconstrained flat-age prior the posterior was asymmetric (old fiducial total $11.39 + 0.09/-0.15$), present in

all five estimators including the purely empirical raw. It is the age–dust–M/L “outshining” degeneracy in the 4-band fit: the low-M $\star$ tail correlates with young age ($r=+0.90$), high dust ($r=-0.58$), and high sSFR ($r=-0.83$) — young+dusty low-M/L solutions reproduce the same broadband points (formally: $\Delta \ln Z = -0.18$ vs an old+passive fit, identical χ^2 ; `scripts/sed_quiescence_check.py`). The KCWI absorption-line spectrum (no emission) rules out the young+dusty branch, so the headline adopts the quiescence-constrained prior, which removes the tail (stat collapses to $+0.07/-0.06$) and the fiducial $\leftrightarrow$ quiescent spread (0.08 dex) is carried as the SFH-prior systematic. See §3.2 and §3.3.6.

- Per-band single-Sérsic parameters (appendix table; `scripts/sersic_parameter_table.py`):

Band	$\lambda_{\text{piv}} (\text{\AA})$	$r_e (")$	$r_e (\text{kpc})$	n	b/a	PA ($^\circ$)	$\mu_e (\text{mag}/''^2)$	$m_{\text{AB}}^{\text{tot}}$
F200LP	4972	2.002 ± 0.122	14.09 ± 0.86	1.24 ± 0.15	$\geq 0.95^\dagger$	$—^\dagger$	25.23	20.92 ± 0.07
F140W	13923	1.765 ± 0.015	12.42 ± 0.11	1.44 ± 0.15	0.86 ± 0.06	4 ± 20	22.36	18.42 ± 0.01
F150W2	16720	1.643 ± 0.005	11.57 ± 0.03	1.22 ± 0.15	0.80 ± 0.06	11 ± 20	21.91	18.29 ± 0.00
F322W2	32470	1.757 ± 0.010	12.37 ± 0.07	1.59 ± 0.15	0.85 ± 0.06	6 ± 20	21.79	17.82 ± 0.01

† F200LP: the deflector is faint in this band, so the single-Sérsic ellipticity/PA are unconstrained (consistent with circular); the IR bands set the shape. Errors are bootstrap, with $b/a/PA/n$ floored by the injection-recovery scatter; the mask-choice systematic (§2.1.1b) dominates and is separate. The deflector is mildly elliptical ($b/a \approx 0.80-0.86$, $PA \approx 4-11^\circ$ E of N) with low Sérsic index ($n \approx 1.2-1.6$) — consistent with the $b/a=0.75$ adopted for the aperture from the independent light fit. - Aperture provenance / why this supersedes 11.36: the old per-band apertures were hand-drawn in the GUI and inconsistent — F200LP $a=1.19''$ ($\sim 0.5 R_e$) captured only 17% of the light vs IR $\sim 46\%$ — so F200LP was under-measured $\rightarrow$ SED artificially red $\rightarrow$ M $\star$ biased HIGH (11.36). The matched aperture measures the same region in every band $\rightarrow$ bluer SED $\rightarrow$ lower M/L $\rightarrow$ the empirical raw drops to 11.22. - Flux uncertainties: the *propagated statistical* errors are 0.03–1.6% (F322W2 0.03%, F150W2 0.05%, F140W 0.4%, F200LP 1.6%) — 6–362 $\times$ below the adopted 10% floor. The 10% is a deliberate systematic floor (zeropoint, aperture, drizzle correlated noise $\sim 1.3-2\times$, SED-model mismatch), not the measurement error — report the true errors and state the floor ($\sim 5\%$ is better-motivated; precedent in GAMA/Taylor+2011, whose error budgets are calibration-dominated, not photon-noise). Floor sensitivity (`results/aperture_floor5_check.npz`): 5% $\leftrightarrow$ 10% shifts every estimator’s central by $\leq +0.02$ dex (headline total 10%=11.50, 5%=11.52, Planck 2015) with tighter stat at 5% — well within the ± 0.12 systematic, so the headline is robust to the floor; 10% kept for the headline. - Superseded values (cross-checks): mismatched-aperture 11.36 (F200LP-biased high); 2-component-mask 11.16 $+0.31$ (over-masked $\rightarrow$ biased low); expert-aperture 11.33. - “Typical elliptical at $z \sim 0.7$ ”: M $\star \approx 1.6 \times 10^{11} M_\odot$ (empirical raw) to $\approx 3.2 \times 10^{11}$ (aperture-corrected total headline) at $z = 0.6756$, passive/quiescent SFH (mass-weighted age ~ 5 Gyr, low sSFR). Place on the $z \approx 0.5-1.0$ mass-size relation (van der Wel et al. 2014; Mowla et al. 2019) and the strong-lens-deflector analog sample (Sonnenfeld et al. 2015, SL2S V — best z -match). [notebook 10 verified citations] - X-ray / quiescence: GAP G2 — no X-ray data in repo. Source externally if you want the “truly quiescent” claim; otherwise frame from the passive SED only.

§3.3 Mass Estimates from Lens Modeling

G1 — sourced from the Ferrami et al. draft (2026-06-12); see §2.3 for method + caveats. Every number below is PROVISIONAL — the draft stamps the evidence table, the M $\bullet$ - σ figure, and the potential-correction figure PLACEHOLDER UNTIL FINAL RUNS ARE COMPLETED. The draft is also internally inconsistent on the headline BH mass (several values coexist, see flag) because the final sampling has not landed. Do not freeze any lens-model number into our draft until Ferrami’s final runs complete. All masses on Ferrami’s Planck-2015 cosmology ($H_0=67.7$, $\Omega_m=0.302$) — convert to our ($H_0=70$, $\Omega_m=0.3$) frame before any joint M $\bullet$ - σ / M $\bullet$ -M $\star$ plot (§2.3 reconciliation flags).

§3.3.1 Headline result

- Canonical BH mass = free-BH combined posterior; uncertainty = 1σ errors across the selected (free-BH) EPL models: $M_\bullet = 5.2_{+1.7/-1.4} \times 10^8 M_\odot$ (range $3.8 - 6.9 \times 10^8$; $\log M_\bullet = 8.72 + 0.12/-0.14$). The central is the model-averaged free-BH posterior (BH position free within $0.5''$; the highest-evidence family in Table

1 — cEPL+free $\Delta \ln Z = 0$, bEPL+free -0.08 , sEPL+free -1.43 ; per-model cEPL 5.51 ± 1.40 , sEPL $5.01_{-1.20/+1.50}$, bEPL $5.11_{-1.30/+1.50} \times 10^8$, agreeing to $\sim 10\%$. The uncertainty is the 1σ model envelope (min median $-\sigma_{\text{lo}}$ to max median $+\sigma_{\text{hi}}$ over the three selected free-BH models) = $3.8\text{--}6.9 \times 10^8$ (user 2026-06-16). The fixed-BH runs (cEPL 1.61, sEPL 2.10, bEPL 1.86×10^9 ; lower evidence, $\Delta \ln Z -3.0$ to -3.65) are NOT in the final selection (kept only as the lower-evidence alternative; the draft prose’s “fixed-BH best model” framing must be reconciled, §3.3.2). Uniform prior $[10^8, 10^{11}] \text{ M}\odot$.

- Provenance / caveat: the central is the free-BH split-normal mixture of the Ferrami Table 1 summary medians $\pm$ errors (scripts/bh_mass_combine.py, seeded; equal-weight 5.22 , evidence-weighted 5.29×10^8). The uncertainty is the free-BH (selected EPL) 1σ model envelope (min median $-\sigma_{\text{lo}}$ to max median $+\sigma_{\text{hi}}$ over the 3 free-BH solutions) = $3.8\text{--}6.9 \times 10^8$. The rigorous version stacks Giovanni’s actual MultiNest chains and resolves free-vs-fixed — flagged TODO; until then the envelope is the honest provisional error.
- Detection significance $> 4\sigma$: models with a central point mass have higher Bayesian evidence than smooth-only (no-BH) models at $> 4\sigma$ — the radial image *requires* a central point-like mass. (The no-BH and extended-mass SIS/NFW perturbers are all disfavoured by $\Delta \ln Z \approx -4$ to -6 ; §3.3.3.)
- $\triangle$ Supersedes the earlier “ $\sim 2 \times 10^9$ fixed-BH” framing. The Ferrami draft *prose* (intro/results + best-model figure caption) is written around the fixed-BH solutions ($\sim 1.6\text{--}2.62 \times 10^9 \text{ M}\odot$), but those have lower evidence than the free-BH runs. We adopt the free-BH combined posterior (5.2×10^8); the fixed-BH values are kept below only as the lower-evidence alternative. Giovanni’s prose still calls sEPL+fixed-PM the “best model” — this must be reconciled (the data, when the BH is free to move, prefer a $\sim 3\text{--}4 \times$ lower mass).

§3.3.2 Best model + why the slope matters

- Highest-evidence model = a FREE point mass on the cEPL (or bEPL) main halo — $\Delta \ln Z = 0$ (cEPL+free) / -0.08 (bEPL+free) in Table 1, BH free within $0.5''$ of the light centre $\rightarrow$ our canonical $M_\bullet = 5.2 \times 10^8$ (§3.3.1). $\triangle$ The draft prose instead designates sEPL + a FIXED point mass as the “best model” (and the best-model figure caption shows 2.62×10^9) — a fixed-vs-free inconsistency: the fixed runs have *lower* evidence ($\Delta \ln Z \approx -3.0$ to -3.65) yet are what the prose/figure quote. We adopt the free-BH result; the final runs / Giovanni’s prose must resolve this. (Evidence is quoted as $\Delta \ln Z$; the placeholder table is not renormalised to any single reference.)
- The main halo is markedly sub-isothermal: $\gamma \approx 1.31 \pm 0.08$ ($\theta_E \approx 1.36''$). A shallow total slope is *what produces the bright radial arc extending to the deflector centre* — the lens environment (it sits ~ 100 kpc from the cluster ACT-CL J0206.2–0114) flattens the projected mass. This is the “shape of the mass profile matters” argument: the radial image only stays bright down to the centre for a shallow slope, and it is the radial image that the BH perturbs. [Ferrami draft §Data, §Mass model comparison]

§3.3.3 Model-comparison table (15 models — PLACEHOLDER, Ferrami draft Table 1)

3 smooth main-halo families (sEPL/cEPL/bEPL) $\times$ 5 perturbers (free BH / fixed BH / free NFW / free SIS / no BH). $\Delta \ln Z$ are as printed in the draft (NOT yet renormalised to the sEPL+fixed-PM reference):

Smooth model	$\Delta \ln Z$	d.o.f.	Perturber posterior
cEPL + free BH (<i>highest evidence</i> $\rightarrow$ <i>canonical family</i>)	0	37	$M_\bullet = 5.51 \pm 1.40 \times 10^8 \text{ M}\odot$
cEPL + fixed BH	-3.25	35	$M_\bullet = 1.61 \pm 0.21 \times 10^9 \text{ M}\odot$
cEPL + free NFW	-4.88	38	$M_{\text{vir}} = 6.48 \times 10^9$, $c_{\text{vir}} = 40$
cEPL + free SIS	-5.26	38	$\theta_E = 0.041''$, $\gamma = 2.59$
cEPL (no BH)	-5.06	34	—
sEPL + free BH (<i>canonical family</i>)	-1.43	36	$M_\bullet = 5.01_{-1.20/+1.50} \times 10^8 \text{ M}\odot$
sEPL + fixed BH (<i>draft prose “best model” — lower evidence</i>)	-3.65	34	$M_\bullet = 2.10^{+0.19}_{-0.30} \times 10^9 \text{ M}\odot$
sEPL + free NFW	-4.76	37	$M_{\text{vir}} = 2.65 \times 10^9$, $c = 91$
sEPL + free SIS	-5.80	37	$\theta_E = 0.027''$, $\gamma = 2.47$
sEPL (no BH)	-4.78	33	—

Smooth model	$\Delta \ln Z$	d.o.f.	Perturber posterior
bEPL + free BH (<i>canonical family</i>)	−0.08	38	$M_{\bullet} = 5.11_{-1.30}^{+1.50} \times 10^8 M_{\odot}$
bEPL + fixed BH	−3.04	36	$M_{\bullet} = 1.86 \times 10^9 M_{\odot}$
bEPL + free NFW	−4.35	39	$M_{\text{vir}} = 5.78 \times 10^9, c = 127$
bEPL + free SIS	−4.82	39	$\theta_E = 0.025'', \gamma = 2.46$
bEPL (no BH)	−5.14	35	—

- Pattern (robust across all 3 halo families): the no-BH and free-SIS/free-NFW (extended central mass) models are all disfavoured by $\Delta \ln Z \approx -4$ to -6 ; a point mass is preferred. The FREE-BH runs carry the HIGHEST evidence ($\Delta \ln Z$ 0 / −0.08 / −1.43) and cluster at $\sim 5 \times 10^8 M_{\odot} \rightarrow$ the canonical combined value (§3.3.1); the *fixed*-BH runs have lower evidence ($\Delta \ln Z$ −3.0 to −3.65) and land ~ 3 – $4\times$ higher (1.6 – 2.1×10^9). Earlier drafts of this fact sheet treated the fixed value as headline — corrected 2026-06-12 to the free-BH combined posterior.

§3.3.4 Best-model parameters (Ferrami draft Table 2 — PLACEHOLDER; headed “cEPL”, FIXED-BH run)

Note: this parameter table is a fixed-BH solution ($M_{\bullet} = 1.61 \times 10^9$, the lower-evidence run); the canonical BH mass is the free-BH combined posterior 5.2×10^8 (§3.3.1). The smooth-halo parameters below (θ_E , γ , shear, light) are essentially common to the free-BH solution — only the point-mass row differs.

Component	Parameter	Posterior
Main deflector	θ_E	$1.360_{-0.08}^{+0.10}''$
	γ (slope)	1.31 ± 0.08 (sub-isothermal)
	r_c (core)	$0.0012''$
	f (axis ratio)	0.826
	φ	10.8°
Secondary deflector (SIS)	θ_E	$0.194''$
Tertiary deflector (SIS)	θ_E	$0.017''$
External shear	Γ	0.095 ± 0.003
	θ_Γ	-14.8°
Point mass (<i>fixed-BH; NOT canonical</i>)	$M_{\bullet}$	$1.61_{-0.21}^{+0.21} \times 10^9 M_{\odot} \rightarrow$ use free-BH 5.2×10^8 (§3.3.1)
Lens light (Sérsic)	R_{eff}	$0.051''$
	n_s	1.372
	f (axis ratio)	0.834
Source light (Delaunay)	λ	46798
	w_a, w_b	1.728, 0.451

(Circular multipole and lens-light multipole coefficients a_m/b_m , $m=1,3,4$ are all $|a|, |b| \lesssim 0.012$ with Gaussian[0,0.01] priors — see draft Table 2 for the full list.)

§3.3.5 $M_{\bullet}$ – σ placement and prior radial-arc BH lenses

- The draft places $M_{\bullet}$ against the local $M_{\bullet}$ – σ relation of van den Bosch (2016, M_{σ} FP), with the only two prior radial-arc / radial-image SMBH lens measurements overplotted: Nightingale et al. ($z = 0.17$) and Melo-Carneiro et al. ($z = 0.45$) — both $\sim 2\sigma$ *above* the local relation. DES J0206–0114 ($z = 0.675$) is the highest-redshift and most centrally-extended of the three.
-  With the canonical $M_{\bullet} = 5.2 \times 10^8$ (free-BH), the placement SENSE flips vs the old $\sim 2 \times 10^9$. Rough KH13 cross-check ($\log M_{\bullet} = 8.49 + 4.38 \cdot \log(\sigma_e/200)$, intrinsic scatter 0.29 dex): at $\sigma_e = 267$ km/s KH13 predicts $M_{\bullet} \approx 1.1 \times 10^9$, so our 5.2×10^8 ($\log M_{\bullet} = 8.72$) sits ≈ 0.3 dex ($\sim 1\sigma$) *BELOW* the local relation — i.e. *slightly under-massive* for its σ , the opposite of the fixed-BH $\sim 2 \times 10^9$ (which would sit *above*, like

Nightingale/Melo-Carneiro). (*This is a labeled rough calc on KH13, NOT the paper result; the actual placement vs vdB16 is Giovanni's figure — the narrative “above the relation” framing in the draft prose needs reconciling with the free-BH value.*)

- The $M_\bullet$ - σ figure is a PLACEHOLDER; the σ comes from our companion paper (Córdova Rosado et al., in prep. — $\sigma_e = 267 \pm 13$ km/s, this repo). The adopted $M_\bullet$ uncertainty is the 1σ errors across the selected free-BH EPL models — the 1σ model envelope $3.8\text{--}6.9 \times 10^8$ ($5.2 -1.4/+1.7$; §3.3.1). The fixed-BH runs ($\rightarrow 2.3 \times 10^9$) are NOT folded in (not in the final selection). This is exactly the 4D ($M_\bullet$, σ , $M_\star$, z) joint constraint PROJECT_BRIEF.md flags — and now that both papers share the Planck-2015 cosmology (§2.3), the join is direct.

§3.3.5a Quantitative offset from the Greene+2020 relations (`scripts/relation_offset_significance.py`)
Rigorous version of the rough KH13 calc above, against the Greene+2020 relations actually drawn in our figures (params from the figure code: $M_\bullet$ - σ $\alpha=8.03$, $\beta=4.24$, $\epsilon_{\text{int}}=0.43$ dex, pivot $\sigma=160$; $M_\bullet$ - $M_\star$ $\alpha=7.89$, $\beta=1.33$, $\epsilon_{\text{int}}=0.65$ dex, pivot $M_\star=3 \times 10^{10}$). With $\log M_\bullet = 8.72$ (PROVISIONAL), $\sigma_e = 267$ km/s, $\log M_\star = 11.46$:

- Distance to the scatter-free (mean) relation: $\Delta \log M_\bullet = -0.26$ dex ($M_\bullet$ - σ) and -0.49 dex ($M_\bullet$ - $M_\star$) — a *mildly under-massive* BH for its σ and $M_\star$ (the same sense as the KH13 cross-check, the opposite of the fixed-BH $\sim 2 \times 10^9$). Against the singular mean line, using our measurement errors only ($\sigma_{M_\bullet} \oplus \beta \cdot \sigma_x$), this is 1.8σ for both relations — i.e. the point is measurably off the *mean* line, but that is the wrong yardstick for a single object (see next).
- Consistency with the relation, the pythagorean way: a single galaxy is drawn from a population with intrinsic scatter ϵ_{int} about the mean, so its consistency is $N_\sigma = |\Delta| / \sqrt{(\sigma_{M_\bullet})^2 + (\beta \cdot \sigma_x)^2 + \epsilon_{\text{int}}^2}$. This gives $N_\sigma = 0.57$ ($M_\bullet$ - σ) and 0.69 ($M_\bullet$ - $M_\star$) ($0.56 / 0.67$ including the relation-parameter errors) — our point falls fully within the intrinsic scatter of both local relations ($< 0.7\sigma$), the scatter term ($0.43 / 0.65$ dex) dominating the budget.
- Framing note (figures). The shaded Greene+2020 band in Figs 3–4 is the relation's 1σ locus as we choose to plot it — it reflects how well the mean relation is determined (zero-point/slope uncertainty), the appropriate band to draw given we cannot know a priori how much of the population scatter applies to any one object. The intrinsic-scatter term ($0.43 / 0.65$ dex) is then carried *separately* in the N_σ above, which is what establishes that AGEL J0206 is consistent with the relations. Both numbers should be quoted: “ $\Delta \log M_\bullet = -0.26 / -0.49$ dex from the mean relation, i.e. within $0.6 / 0.7\sigma$ once the relation's intrinsic scatter is included.”
- *Caveat*: M_{BH} is PROVISIONAL (Ferrami, free-BH 1σ envelope); these track it. The registry leaf `mbh_sigma_offset` (-0.69) was computed against $[\text{mean} + \epsilon_{\text{int}}]$ (scatter added to the mean) and is not the mean-relation offset (-0.26) used here — to be reconciled.

§3.3.6 Quiescence evidence (also fills gap G2 — X-ray / radio)

The Ferrami draft §Data carries the multiwavelength quiescence argument that was gap G2 in this repo (no longer external-only):

- No AGN emission lines in the KCWI IFU central spectrum (Córdova Rosado et al., in prep.).
- X-ray: no Chandra source at the galaxy position; $0.5\text{--}7$ keV flux upper limit 1.8×10^{-15} erg s $^{-1}$ cm $^{-2}$ (Chandra DR2; field obs PI Hughes, ID 16229, 30 ks) $\rightarrow L_X < 10^{43}$ erg s $^{-1} \approx 10^{-4} \times L_{\text{Edd}}$ (the draft uses a $10^9 M_\odot$ BH; for our canonical $5.2 \times 10^8 M_\odot$, L_{Edd} halves $\rightarrow \approx 2 \times 10^{-4} L_{\text{Edd}}$, still order 10^{-4} — the quiescence conclusion is unchanged).
- Radio-quiet: below the LOFAR DR3 detection limit.
- $\rightarrow$ The central gravitational perturbation is from a quiescent (non-accreting) SMBH — the headline novelty (a dynamical-equivalent mass for a quiescent BH at cosmological distance). G2 is now partly internal, but the X-ray flux limit and the Chandra/LOFAR program details should still be cross-checked against the actual Chandra DR2 / LOFAR DR3 sources before we assert them in *our* paper.

§3.4 Structural nature of the deflector: classical elliptical, (un)cored, and the pseudobulge question

Synthesis of the morphology/structural work (notebook 18; `scripts/{sersic_parameter_table, psf_star_census, build_psf_models, core_sersic_test}.py`; TESTS A5f/A6/A7; REPORT `cored_vs_coreless_2026-06-17.md`; kinematics `results/radial_sigma_profile.npz`, `nb11_perbin_perspaxel_wide.npz`).

(a) Overall class — a massive, quiescent early-type galaxy (robust). Multiple independent lines agree: $\sigma_e = 267$ km/s; $\log(M_\star/M_\odot) \approx 11.46$; a pure old-stellar-population absorption spectrum (strong Ca H&K, G-band, Mg b, Fe; no emission lines — the only emission is the $z=1.302$ lensed source, masked); passive SED (mass-weighted age 3.8 Gyr, $A_V \approx 0.2$, $sSFR \approx -10.5$); smooth single-Sérsic light, mildly flattened ($b/a \approx 0.80-0.86$, $PA \approx 4-11^\circ$) with an extended envelope; in a BCG-like cluster environment (near ACT-CL J0206). Caveat: we cannot cleanly separate elliptical (E) from lenticular (S0) — the low single-Sérsic n (below) and the lack of a clean λ_R map preclude a formal morphological E/S0 split; the giant-elliptical/cD reading is *supported* by σ + the envelope + the environment. For the $M_\bullet-\sigma$ / $M_\bullet-M_\star$ comparison this is immaterial: Greene+2020 calibrate on the combined E+S0 (“early-type”) sample, which is exactly this object’s class.

(b) Classical bulge / elliptical — NOT a pseudobulge (this matters for $M_\bullet-\sigma$ validity). The KH13 scaling relations are calibrated on classical bulges + ellipticals; pseudobulges scatter off $M_\bullet-\sigma$ and are excluded (Kormendy & Ho 2013). On the standard classical-vs-pseudo discriminants (Kormendy & Ho 2013; Fisher & Drory 2008, 2016), our deflector is classical on every axis but one: - Mass: $\log M_\star \approx 11.46$ — far above the (low-mass) pseudobulge regime. - σ : 267 km/s — pseudobulges are low- σ systems. - Dispersion support: within R_e the ordered rotation is low — $V \approx 40$ km/s vs $\sigma \approx 267 \rightarrow V/\sigma \approx 0.15$ (`radial_sigma_combined_posterior.npz`), i.e. slow-rotator / dispersion-dominated, *not* the rotation-dominated ($V/\sigma \gtrsim 1$) pseudobulge. *Caveat*: the KCWI resolution + arc contamination make this a marginal rotation measurement (the outer annular $V(R)$ is contaminated and not reliable) — it supports, but does not by itself prove, dispersion support. - Star formation: passive, no emission, low dust — no disk / ongoing SF / nuclear spiral or bar signatures (pseudobulges are gas-rich, star-forming, diskly). - The one ambiguous feature — Sérsic index. The single-Sérsic $n \approx 1.2-1.6$ (A5f) is *below* the $n \approx 2$ pseudobulge threshold (Fisher & Drory 2008 — low- n bulges also show higher V/σ and lower σ , the opposite of this object). But this is a box-limited artifact: Test A shows n (and the total light) keep rising with fit box and never plateau (F140W n 1.2 $\rightarrow$ 2.0 over box 4 $\rightarrow$ 10”), i.e. the single Sérsic under-counts the extended high- n / cD envelope. The low n is therefore *not* evidence for a diskly pseudobulge — it reflects the fitting box, not the profile. - Verdict: classical bulge / elliptical, established by mass + σ + dispersion support + passivity; the low single-Sérsic n does not overturn it. ☐ AGEL J0206 is a valid, clean probe of the $M_\bullet-\sigma$ / $M_\bullet-M_\star$ relations (not a pseudobulge contaminant).

(c) Cored vs coreless — expected cored, but observationally UNRESOLVABLE (A7). By the core/coreless dichotomy (Faber et al. 1997; Lauer et al. 2007 [ApJ 664, 226, “bimodal central surface-brightness profiles” — NOT the same-year $M_\bullet-\sigma$ selection-bias Lauer+2007]; Kormendy et al. 2009) a $\sigma \approx 267$ km/s ($M_\bullet \sim 5 \times 10^8 M_\odot$) galaxy is *expected* to host a central surface-brightness deficit (depleted core from binary-SMBH scouring). But the expected break radius $r_b \sim 0.02-0.1$ kpc $\approx 3-14$ mas at $z=0.676$ — below every PSF we can build (finest F150W2 FWHM 0.049”, HWHM 0.024” = 0.17 kpc; PSF models for all 4 bands, A6). PSF-convolved core-Sérsic vs single-Sérsic fits (A7, `core_sersic_test.py`, F150W2 + F140W): a free core-Sérsic is “preferred” by $\Delta BIC = 386/43$, but this is a fitting artifact, not a detection — the fitted $r_b = 2.5-3.6$ kpc is 10–50 $\times$ too large for a depleted core and the single-Sérsic inner residual does *not* deepen toward $r=0$; the core-Sérsic is absorbing the global single-Sérsic / cD-envelope mismatch (same box-limitation as in (b)), not a nucleus. Verdict: cored vs coreless is indeterminate (upper limit $r_b < 0.024'' = 0.17$ kpc); a genuine depleted core removes $<1\%$ of the light $\rightarrow$ negligible for $M_\star$, and the $M_\bullet$ -core-scouring cross-check is not feasible with these data. The single-Sérsic remains the pragmatic *inner* light model; the structure that *does* matter for $M_\star$ is the outer envelope (Test A), already carried in the $M_\star$ systematic budget (§3.2). Resolving the core would require $\sim$ mas imaging (ELT-class) — out of scope, and the science payoff ($<1\%$ -light core) does not warrant it.

Key citations (verified in notebook 10)

- ppxf: Cappellari & Emsellem (2004); Cappellari (2017, MNRAS 466, 798); Cappellari (2023).
- σ_e definition: Gültekin et al. (2009) eq. 1 (the luminosity-weighted second-moment integral — verified 2026-06-12); co-add-then-fit method Cappellari et al. (2006, MNRAS 366, 1126, SAURON IV §2.3) [NB: that paper's eq. 1 is the *aperture correction*, NOT σ_e — do not cite “Cappellari 2006 eq. 1” for σ_e]; the discrete spaxel/bin sum (§7) is Cappellari et al. (2013, ATLAS3D XV, MNRAS 432, 1709) eq. 29 (= the discretised Gültekin 2009 eq. 1; eq. 29 also has an inclination/projection term that our observed-plane sum omits) [verified 2026-06-12]; Kormendy & Ho (2013, ARA&A 51, 511) eq. 3 (scatter 0.29 dex); Greene, Strader & Ho (2020, ARA&A 58, 257) fig. 5 [$\triangle$ AUDIT FLAG — see below]; Saglia et al. (2016).
- z=0 IFS calibration: Cappellari et al. (2013, ATLAS3D XV & XX); Krajnović et al. (2013, XVII); Zhu et al. (2024, MaNGA DynPop III).
- Mass-size / analogs: van der Wel et al. (2014); Mowla et al. (2019); Ge et al. (2021); Sonnenfeld et al. (2015, SL2S V) [$\triangle$ AUDIT FLAG — see below].
- Vacuum $\leftrightarrow$ air: Ciddor (1996). Wide-window precedent: Bezanson et al. (2018); van der Wel et al. (2021).
- Source-z catalog: AGEL DR2 (Carleton et al., in prep). Cluster: Hilton et al. (2021, ACT DR5).

Citation audit + referee challenges (2026-06-08)

Citation audit (vs the user's Zotero library; “not in library” $\neq$ fabricated — most are real foundational papers simply not yet added to Zotero, flagged so the .bib can be completed):

- Verified in-library, metadata exact: Cappellari (2017, MNRAS 466, 798) ✓; Kormendy & Ho (2013, ARA&A 51, 511) ✓; Greene, Strader & Ho (2020, ARA&A 58, 257) ✓ (*metadata only — see flag*); Hilton et al. (2021, ApJS 253, 3 = ACT SZ-cluster catalog) ✓; Cappellari (2023, MNRAS 526, 3273) ✓. Also in-library and relevant: McConnell & Ma (2013) — a massive-BH $M_\bullet - \sigma$ / $M_\bullet - M_{\text{bulge}}$ anchor.
- $\triangle$ FLAG 1 — Greene, Strader & Ho (2020, ARA&A 58, 257): bibliographic metadata is CORRECT, but this review is titled “Intermediate-Mass Black Holes.” It does provide $M_\bullet - \sigma$ / $M_\bullet - M_\star$ relations (legitimately citable as a *comparison* relation, as Fig. 4 uses it), but for a $\sim 10^9 M_\odot$ SMBH the *primary* anchors should be KH13 / McConnell & Ma 2013 / Saglia+2016, with Greene+20 as one of the overplotted comparison relations. ACTION: verify the “fig. 5” number (the scaling-relation figure in Greene+20 may not be fig. 5) and ensure the text frames it as a comparison, not the primary massive-BH relation. (*Not a fabrication — a framing/figure-number check.*)
- $\triangle$ FLAG 2 — Sonnenfeld “SL2S V” (2015): the library contains SL2S IV (Sonnenfeld et al. 2013, ApJ 777, 98), not paper V (2015, ApJ 800, 94). ACTION: decide which you mean and either correct to “IV (2013)” or add the 2015 V paper to Zotero. Do not let the 2013 record stand in for a 2015 citation.
- $\triangle$ FLAG 3 — Bezanson et al. (2018): no Bezanson-first-author 2018 item in the library. If you mean the LEGA-C velocity-dispersion paper (Bezanson et al. 2018, ApJ 858, 60), add it / verify.
- To add to Zotero before the .bib is complete (real papers, just absent; metadata NOT independently verified here — confirm on add): Cappellari & Emsellem (2004), Cappellari et al. (2006, SAURON IV), Gültekin et al. (2009), Saglia et al. (2016), Cappellari et al. (2013, ATLAS3D), Krajnović et al. (2013), Zhu et al. (2024, MaNGA DynPop III), van der Wel et al. (2014), Mowla et al. (2019), Ge et al. (2021), Ciddor (1996), van der Wel et al. (2021, LEGA-C DR3).

Referee challenges to the current doc (anticipate / address in drafting):

1. R_e method family (was “non-converged”; largely ADDRESSED 2026-06-11). Headline $R_e = 2.097''$ (best-mask raw CoG, $r_{\text{max}}=6''$) is now photutils-validated to $\pm 0.002''$ (`validate_Re_photutils.py`: independent RadialProfile integration), and the raw CoG sits at the top of a tight best-mask method family — sky-sub CoG $1.922''$, Sérsic $r_{\text{eff}} 1.897'' \rightarrow$ method systematic $\pm 0.100''$ carried explicitly. The raw-CoG-is-top-of-family caveat remains (no sky pedestal subtraction), but it is now bounded and documented, not a flagged unknown. Knock-on: R_e sets the σ_e aperture and (weakly) $M_\star$; both effects are inside their systematic budgets.
2. R_e -source budget term (RESOLVED 2026-06-11). Now uses the best-mask CoG light family only

{1.912/2.097/2.281"} $\rightarrow \pm 5.13$ — the referee-preferred light-family treatment. CaHK+G 2.90" (a different I-map definition) and the full grid (± 9.98) are reported as cross-checks, NOT folded, so the budget no longer conflates aperture-uncertainty with $\sigma(R)$ -physics.

3. Rising outer $\sigma(r)$ attributed to arc dilution. The claim " $\sigma(r)$ should decrease; any rise is the $z=1.302$ arc" is the right physical call, but it is *also* what lets us drop the high- σ outer bins; make the arc-contamination evidence explicit (EW / continuum-dilution diagnostic, nb07e) so it does not read as motivated reasoning.
4. He I 3819 source-emission mask rests on a 3-pixel residual cluster — defensible (consistent across reductions, matches $z=1.302$), but small; a referee may probe it. The M8/M9/M10 audit trail covers this.
5. $\sigma_{\text{inst}} = 12.6$ km/s vs $\sigma_e \approx 270$: well-resolved, not a challenge — but state the LSF treatment.

Flagged open items (carry into drafting)

1. ~~R_e-source systematic not folded in~~ RESOLVED 2026-06-08 (M12, nb15): re-measured at the wide window (± 6.13) and folded into the budget $\rightarrow$ headline $\pm 11.77 \rightarrow \pm 13.27$.
2. Reduction-pass ± 3.45 rests on only 2 reductions — refine if a 3rd lands.
3. ~~H δ may still need targeted masking~~ RESOLVED 2026-06-08 (M12, nb16): decision = keep unmasked (H δ is well-fit, not a contaminant; masking it would discard LOSVD information). TODO in `bootstrap_ppxf.py` closed. 3b. NEW (A3c, nb14): raw CoG R_e is `r_max`-non-convergent; headline 2.305" is the top of a 2.1–2.5" family (R_e method systematic ± 0.08 "), not folded into $\sigma_e/M\star$ — flagged for a decision.
4. Asymmetric-error tables: older 4-corner/M10 tables print $-17.92/+17.65$ (pre-M11), M11 was $-11.98/+11.57$ (± 11.77); current (M12) is $-13.45/+13.10$ (± 13.27). Quote the M12 value (canonical: `results/PAPER_VALUES.json`).
5. No X-ray / quiescent data in repo (G2). 6. No lens-model content in repo (G1).
6. "We do not account for peculiar velocities" needs to be written in (G3 — true, just add it).
7. Source- z (1.302) error bar + its cache are a [TBD] placeholder.
8. K409 (Aug-30) PI and per-night DIMM seeing still TBD for acknowledgements.

Internal appendix — σ_e measurement audit trail (M8→M12) — NOT for the manuscript

Superseded by "best mask throughout" (2026-06-11). The numbers in THIS appendix ($\sigma_e=269.62$, total ± 13.27 , R_e-source ± 6.13 , R_e=2.305") are the M12 (2026-06-08) state. They were the headline until the best-mask cascade adopted R_e=2.097" (best-mask CoG, photutils-validated), which moved $\sigma_e \rightarrow 267.31 \pm 11.77$ and re-derived R_e-source $\rightarrow \pm 5.13$ (best-mask CoG family; `run_sigma_e_Re_grid.py`). The current state is §2.1.2 / §2.4.3 / §3.1 above. This trail is kept verbatim as M12 provenance; do not quote its numbers as current.

This is the iterative history of how the final §2.4.3 masking + systematic-component set was reached. The Methods section should present the final procedure (§2.4) and the budget (§3.1), not this trail. Kept here as internal provenance / referee-rebuttal backup. Each step is also a TESTS row with its own cache.

- M8 — He I 3819 (added as a source-emission mask). A 3-pixel positive-residual cluster at def-rest 5244–5248 Å = source-rest 3818–3820 Å matches He I 3819.6 from the $z=1.302$ source; consistent across both reductions $\rightarrow$ astrophysical, not a CR. Masked. σ_e (NEW) 268.98 $\rightarrow$ 271.87.
- M9 — 5193–5204 Å bump (deliberately NOT masked). A +5–7% feature; checked against the arc spectrum (no source counterpart) and the noise spectrum (no sky counterpart) $\rightarrow$ it is the Mg b LOSVD wing (signal). Smoke-masking it drops σ_e by 7 km/s $\rightarrow$ DO NOT mask. (Same "it's signal, not contamination" logic later applied to H δ in M12.)
- M10 — full sky-line audit. Flag every band with `cube noise_sky > 2.5× median` across the fit window, and cross-check each against the arc spectrum to confirm no source counterpart $\rightarrow$ 9 OH/sky bands added to `BAD_PIXELS_REST` (26 $\rightarrow$ 35 entries). σ_e (NEW) 271.87 $\rightarrow$ 269.62.

- M11 — cube-matched systematic re-derivation. Re-ran the I-shape / mask-weight / fit-window sweeps on the NEW cube + M10 masks (replacing carried OLD-cube values). The fit-window term collapsed $\pm 15 \rightarrow \pm 3.82$ on the cleaned cube; total sys $\pm 17.16 \rightarrow \pm 10.81$ (total ± 11.77).
- M12 (2026-06-08) — R_e-source fold-in + two cross-checks. Folded the wide-window R_e-source systematic (± 6.13) into the budget $\rightarrow$ sys ± 12.43 , total ± 13.27 . Added the arc-mask-definition cross-check (± 4.58 ; overlaps the F200-mask term, not double-counted) and resolved the H δ open item (keep unmasked — well-fit, masking it discards LOSVD information). Central σ_e 269.62 unchanged across M10 $\rightarrow$ M12; only the error grew.

Net effect of the trail: central σ_e settled at 269.62 km/s by M10 and has not moved since; the error bar evolved ± 17.78 (pre-M11) $\rightarrow \pm 11.77$ (M11) $\rightarrow \pm 13.27$ (M12). The mask set (He I + 35-entry sky/CR + no-Balmer) and the 7-component budget that the final §2.4 procedure uses are the M12 result.